



# The cost of innovation: R&D and high cash holdings in U.S. firms<sup>☆</sup>



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## ABSTRACT

We show that R&D investment explains a significant portion of the increase in the average cash-to-assets ratio of U.S. firms, which more than doubled between 1980 and 2012. In 1980, an average firm held \$0.04 in cash for \$1.00 of R&D spending, but this had increased to \$0.60 by 2012. The increasing sensitivity of cash holdings to R&D investment and the increase in R&D spending of the typical firm explain over 20% of the increase in aggregate cash holdings. Intensified domestic and global competition appears to be an important explanation for the increased propensity of R&D-intensive firms to hoard cash.

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## 1. Introduction

R&D driven innovation has long been regarded as a crucial driver of productivity and economic growth (Solow, 1957). Although a large body of research has examined the factors that affect the incentives to innovate (e.g., Manso, 2011; Aghion et al., 2013; Aggarwal and Hsu, 2014; Atanassov, 2015, among several others), much less is known about the effect of corporate innovation on the aggregate financing of U.S. firms. Even less is known about how the relationship between R&D and firm financing across U.S. firms may have changed or evolved over time.

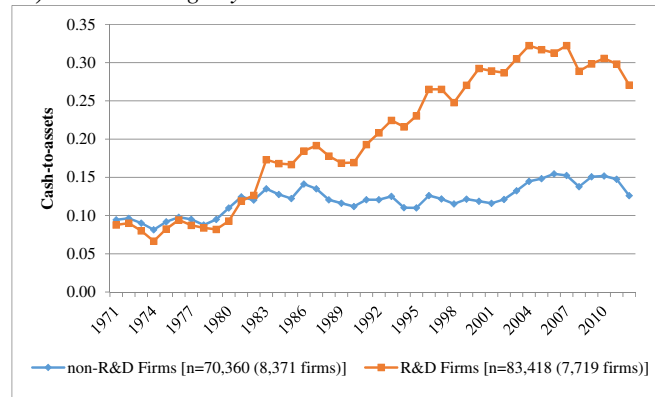
Our study is thus motivated by three recent empirical regularities with respect to R&D and cash holding. First, aggregate cash holdings have increased significantly over the past three decades. By 2012, industrial firms in the U.S. held \$1.5 trillion in cash, up from \$265 billion in 1981, a fact that has attracted unprecedented attention from the business press, financial economists and government. Second, a related and concurrent phenomenon is that the average firm's cash-to-assets ratio has also increased

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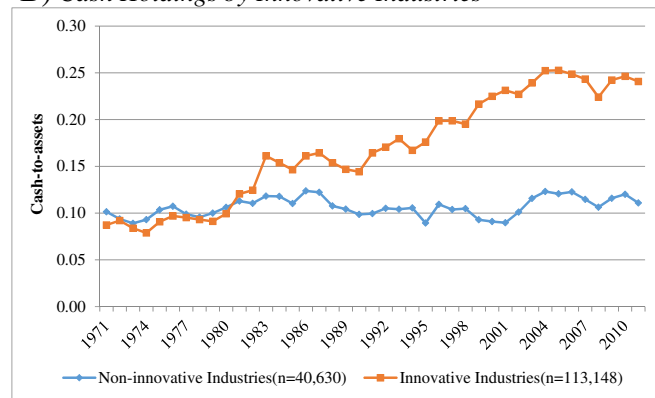
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## A) Cash Holdings by R&amp;D Firms



## B) Cash Holdings by Innovative Industries



**Fig. 1.** The secular trend in the cash-to-assets ratio of R&D and Non-R&D firms from 1971 to 2012. This figure shows the secular trend of average cash-to-assets ratios for R&D firms (Panel A) or firms in innovative industries (Panel B) over the 1971–2012 period. R&D firms are defined as firms with non-zero R&D stock in a given period. An industry is innovative if the average R&D-to-assets for the industry during the preceding year is greater than the median average R&D-to-assets across all industries. The sample includes all publicly traded non-financial and non-utility U.S. industrial firms with positive book assets, positive sales, non-negative common equity, and non-missing values for cash and short term equivalents in Compustat, a total of 153,778 firm-year observations for 15,052 unique firms. The number of firm-year observations is reported in parentheses.

dramatically, and has tripled since 1970. This increase in the cash-to-assets ratio is not limited to large firms (as is sometimes assumed in the popular press) but is pervasive across all size quintiles (Bates et al., 2009). Third, as Fig. 1 (Panel A) clearly shows, the secular increase in the cash-to-assets ratio is largely present only among R&D-intensive firms whose cash-to-assets ratios have nearly tripled since 1980, in sharp contrast to the stable cash-to-assets ratios of firms without R&D over the same period. This third empirical regularity, in particular, has largely been unnoticed or ignored by prior literature. As such the broad goal of our study is to systematically examine how, and to what extent, R&D investment explain changes in the cash holding behavior of U.S. firms in the past three decades.

In principle, firms should hold cash as required by their operating and competitive circumstances. In other words, firm characteristics determine, to a significant extent, the optimal level of a firm's cash holdings (Opler et al., 1999). This suggests that there are two potentially straightforward (though not mutually exclusive) explanations for the aggregate increase in cash holdings. The first potential explanation is that the characteristics of the typical firm have changed significantly over time. This idea, which has been explored most notably by Bates et al. (2009), contends that those characteristics that have positive (negative) effects on cash holdings have increased (decreased) significantly.<sup>1</sup>

The second potential explanation, which has received far less attention in the literature, is that the *sensitivity* of firm cash holdings to firm characteristics may have changed over time, such that even if these characteristics had remained the same, the level of cash holdings would still have increased. While we do not dispute the fact that changing firm characteristics account for a significant part of the explanation for the growth in aggregate cash holdings, we posit that the sensitivity of firm cash holdings to firm factors may have itself changed since 1980, and that this *changing sensitivity* may be as important as *changing firm characteristics* to explain this cash hoarding phenomenon.

<sup>1</sup> Specifically, Bates, Kahle, and Stulz (2009) suggest that the aggregate cash holdings across the U.S. have increased because the typical firm has more volatile cash flows, is more R&D-intensive, and holds smaller net working capital (NWC) than in the past.

Specifically, in this paper, we argue that a significant portion of the secular increase in cash holdings can be explained by a fundamental shift in the sensitivity of cash holdings to R&D investment – a shift in the demand curve for cash of firms engaging in innovation. We show that many of the driving elements of cash holdings explored in the literature interact with, and are intensified by, the effect of R&D intensity on cash balances. This evidence implies that the fundamental association between R&D and cash holding behavior itself has changed significantly over the past three decades, which has had a significant impact on the cash policies of U.S. firms. Indeed, we find that the increasing sensitivity of firm cash holdings to R&D, coupled with the rising R&D investment since 1980, suggest that R&D *by itself* is able to explain more than 20% of the increase in aggregate cash holdings of U.S. firms over 1980–2012. This impact is non-trivial; compared with other important cross-sectional determinants of cash holdings, R&D has been a particularly important factor in explaining the buildup of corporate cash reserves.

We argue that there are several factors may be contributing to the increasing sensitivity of cash holdings to R&D investment. It is well established that R&D driven innovation is intrinsically difficult to finance using external funds (especially debt) due to its intangible nature, uncertain outcome, and potentially severe asymmetric information problems (Arrow, 1962). Thus, R&D is largely financed from internal cash flow and stock issues (Brown et al., 2009), both of which have become more volatile in the past several decades (e.g., Irvine and Pontiff, 2009). In addition, the large adjustment costs of R&D expenditures cause financial constraints to be more binding on R&D intensive firms, leading these firms to rely more on precautionary cash to smooth R&D (Brown and Petersen, 2011). Furthermore, and perhaps most importantly, domestic and international competition, which has increased across many sectors of the U.S. economy in the past several decades, has increased disproportionately in R&D intensive industries, and may be motivating these firms to either use cash as a strategic weapon (Frésard, 2010; Lyandres and Palazzo, 2012), or hold as a precaution to ensure survival.

To systematically investigate the impact of R&D intensity on the trend of cash holdings in the U.S., we assemble a sample of 109,285 firm-year observations for 13,249 non-financial and non-utility U.S. incorporated firms between 1980 and 2012. Since the bulk of our analysis compares cash holdings of R&D firms with that of non-R&D firms, we divide our sample into two groups. R&D firms are defined as firms with non-zero capitalized R&D stock in any given year, assuming a 15% depreciation rate; other firms are classified as non-R&D firms.<sup>2</sup>

We start by examining the univariate change in the mean cash-to-assets ratios of R&D and non-R&D firms over the sample period. In 1981, both R&D and non-R&D firms had similar cash-to-assets ratios of about 10%. By 2012, the average cash-to-assets ratio of R&D firms had grown almost three-fold to 25.4%, while that of non-R&D firms had only increased marginally to 11.3%. Thus, starting in 1981, there was a statistically and economically significant divergence in the cash holding behavior of non-R&D and R&D firms that continued to widen in the ensuing decades. Even after controlling (in multivariate regressions) for other firm and industry characteristics that have been shown to be correlated to cash holdings, the difference in the cash-to-assets ratios of R&D and non-R&D firms, which was virtually indistinguishable in 1981, had increased to over five percentage points by the end of 2012.

We next find that this divergence in cash holdings of R&D and non-R&D firms has been driven to a significant extent by the notable increase in the sensitivity of cash holdings to R&D investment. After controlling for a host of potential determinants of cash holdings, we observe that, in 1980, the average firm held four cents in cash for every dollar of R&D investment, but this had increased *fifteen-fold* to sixty cents by 2012. In sharp contrast, the sensitivity of cash holdings to other important factors such as dividend payouts, net working capital, and industry cash flow volatility, changed very little over the same period.

We attempt to understand the structural factors that may have strengthened the propensity of R&D-intensive firms to hold cash. In particular, we focus on the role of domestic and international competition. Using several measures of competition, we provide strong evidence that domestic competition has increased more dramatically in R&D intensive industries than it has in non-R&D industries since 1980.<sup>3</sup> The analyses show that the cash-to-assets ratio was significantly and positively associated with the level of industry competition in R&D firms; and the effect of competition on the cash holdings of R&D firms has intensified and generally became stronger between 1980 and 2012. In stark contrast, competition was never a significant determinant of cash holdings in non-R&D firms over the same period. Taken together, the results suggest that not only has competition increased substantially in R&D intensive industries, but firms in these industries have also become more sensitive to the effect of competition on their cash holdings policies.

It is of course possible that cash holdings, R&D investment and domestic product market competition are all endogenous. Large cash reserves enable a firm to challenge rivals' bottom lines by spending more on R&D, thus intensifying competition. This increase in competition may in turn further drive up cash as firms in R&D-intensive industries strategically increase precautionary liquidity buffer, thus leading the average cash balances of these firms to spiral upwards.

To address such source of potential endogeneity in the R&D/competition/cash holdings relationship, we use large reductions in industry-level import tariff rates that occurred between 1972 and 2005 as exogenous shocks to competition, as suggested by Frésard (2010), and Frésard and Valta (2014). Large reductions of import tariff rates trigger a sudden increase in the competitiveness of an industry by facilitating the entry of foreign rivals, and are exogenous to internal corporate policies. Using propensity

<sup>2</sup> Our analysis through the paper is robust to a simpler definition of R&D firm, where an R&D firm is defined as a firm reporting R&D expenditures in any given year. In unreported tests, we have also repeated all the analyses with patent-based metrics to proxy for innovative activity, using data from the NBER Patent Citations Data Project (Hall, Jaffe, and Trajtenberg, 2001). The results are quite in line with the main findings documented here.

<sup>3</sup> These include the standard Herfindahl–Hirschman Index (HHI) industry concentration measure based on the three-digit SIC code, a fitted HHI measure developed by Hoberg and Phillips (2010a), a text-based (TNIC) HHI (Hoberg and Phillips, 2010a, 2010b), and the product market fluidity measure of Hoberg, Phillips, Prabhala (2014).

score matched samples and difference-in-differences regressions around these events, we show that treated R&D firms significantly increased their cash holdings relative to matched firms as a response to the sudden increase in entry threats, while treated non-R&D firms did not. Thus, exogenous changes in competition have had a significant effect on the cash holdings policies of R&D firms, but not on non-R&D firms.

Our study contributes to the empirical literature on cash holdings by identifying the first-order effect on the secular increase in cash holdings, and quantifying how much R&D investment accounts for the recent cash buildup. Our findings suggest that the role of R&D in explaining the cash holding behavior of U.S. firms may have been significantly underestimated. R&D explains *not only* a significant amount of the cross-sectional differences in the cash holdings policies across firms, but appears to have had a paramount role in driving up corporate cash over the last three decades. In particular, our study is the first to conduct a detailed time-series analysis of the association between firm R&D investment and cash holdings.

Moreover, our paper complements and extends the work by Bates et al. (2009), and Falato et al. (2013). The former presents evidence that the typical firm has become riskier and has increased R&D spending over time, while the latter shows that firms are increasingly reliant on intangible capital. Both suggest that *changing firm characteristics* may have led to an increase in firms' cash holdings. We show that another way that R&D, a form of intangible capital, has affected corporate is from the fact that the demand for cash among research intensive firms has steadily increased over time.

Our paper also adds to the literature on the financing of R&D and the impact of R&D smoothing on firms' capital structure. Brown et al. (2009) conclude that R&D is mostly financed by volatile sources: cash flow and stock issues. Brown and Petersen (2011) examine the implications of the increase in R&D investment for liquidity management, and find strong evidence that firms depend heavily on cash reserves to smooth R&D in U.S. manufacturing. Further, Ball et al. (2012) conclude that R&D is far more prone to constraints than capital expenditures, thus pointing out the challenges of funding innovation. Our paper presents evidence consistent with these views. The increased competitiveness in R&D intensive industries makes firms in these industries more sensitive to the volatility of financing. R&D intensive firms have thus had increasingly strong incentives to maintain a liquidity buffer in order to smooth the path of R&D, thereby avoiding the potential high adjustment costs and funding shortages. This shift in the way of how R&D is financed has ultimately contributed to the significant increase in aggregate cash holdings across U.S. firms.

Finally, this study deepens our understanding of the relation between product market competition and firms' cash policies. Despite the large body of literature on cash holdings, the empirical evidence on how market competition affects cash holding decisions is still limited. Using large reductions in tariff rates as a quasi-natural experiment, our study identifies that firms in competitive industries do hold more cash, but this is mostly observable in competitive industries with R&D intensive firms. Competition appears to play less of a role in the cash holding behavior of firms in industries that are not research intensive.

## 2. Data

### 2.1. Sample construction

Our base sample starts with all U.S. incorporated industrial firms drawn from the Compustat North America Fundamentals Annual files from 1980–2012. Industrial firms are defined as companies with SIC codes outside the ranges 4900–4949 (utilities) and 6000–6999 (financials). We require firms to have positive assets and sales to be retained in the sample (as does Bates et al., 2009). We exclude firms with negative common equity values or missing cash and short term investments. Further data requirements for all control variables yield a sample of 13,249 surviving and non-surviving firms, with a total of 109,285 firm-year observations. As mentioned, we divide firms into “R&D Firms” and “Non-R&D Firms” based on whether the firm has positive R&D stock in a given period. R&D stock from all past R&D expenditures is calculated using the perpetual inventory method (Hall et al., 2005).<sup>4</sup>

Our analysis employs the most traditional measure of cash holdings that has been used in prior literature: the ratio of cash to book value of assets, where cash is defined as cash and short term investments. Using alternative measures of cash holdings (e.g., the ratio of cash to market value of assets, or cash to assets net of cash) does not affect all the results in any material way.<sup>5</sup>

There are several key variables used in our empirical analysis. *R&D\_Dummy* is an indicator variable that equals one for “R&D Firms”, and zero otherwise. To measure R&D intensity, we calculate the ratio R&D expenditures to book value of assets, denoted as *R&D-to-assets*. For firms with missing R&D, the R&D is set to zero because these firms do not have material R&D expenses in their 10-K filings (Brown and Petersen, 2011; Hirschey et al., 2012). *Industry Cash Flow (Equity Issuance) Volatility* is defined as the standard deviation of cash flow (equity issues) over assets for the previous ten (five) years, and averaged each year at the two-digit SIC code level. *Firm Equity Issuance Volatility* is the standard deviation of equity issues over assets using a five-year rolling window.

Our primary measure of domestic competition is the Herfindahl-Hirschman Index (HHI), a measure of industry concentration; *Compustat HHI* is computed as the sum of squared market shares based on firms' sales in Compustat at the three-digit SIC code level (See Giroud and Mueller (2010)). A higher value of HHI suggests a less competitive industry. We also employ three other

<sup>4</sup> The perpetual inventory method is illustrated as follows:  $G_{it} = (1 - \delta_{R\&D})G_{it-1} + R\&D_{it}$ , where  $t$  is the year index,  $G_{it}$  is the R&D stock in year  $t$ ,  $R\&D_{it}$  denotes the (\$2012 real dollars) R&D expenses during year  $t$ , and  $\delta_{R\&D}$  is the depreciation rate, which is set to be 15%. The initial stock equals the R&D expenses in the first year divided by the depreciation rate.

<sup>5</sup> In unreported results, we replicate our main analysis using the ratio of cash to market value of assets and find that our inference is unchanged. A full set of robustness test results is available upon request.

measures: *Fitted HHI*, *Text-based (TNIC) HHI*, and *Product Market Fluidity*; these data are obtained from Hoberg and Phillips library.<sup>6</sup> Our proxy for foreign competition is calculated as total value of imports divided by the value of shipments minus value of exports plus value of imports at the four-digit SIC code level for manufacturing industries over 1980–2005. The data is obtained from Peter Schott's website (Schott (2008)).

In our regression analysis, we include a set of firm characteristics that may be correlated with a firm's level of cash holdings as controls (as suggested by Bates et al. (2009)). These control variables, as well as variable definitions and Compustat data item codes are described in Appendix A. To mitigate the effect of outliers on our findings, all variables are winsorized at the 1st and 99th percentiles. All nominal values are converted into CPI-adjusted 2012 dollar values.

## 2.2. Descriptive statistics

Panel A of Table 1 displays summary statistics for all the variables for the full sample. The sample exhibits characteristics comparable to those in prior studies. The mean cash-to-assets ratio is 0.16 with a median of 0.08, similar to those reported in Bates et al. (2009). The average industry cash flow volatility, equity issuance volatility at the firm-level and industry-level are 0.11, 0.10, and 0.11, respectively. For our proxies of product market competition, Compustat HHI has a mean value of 0.23, which is greater than the mean value of Fitted HHI (0.06) but similar to the mean value of TNIC HHI (0.22), while the product market fluidity has an average of 6.71; these are all similar to the values reported by Hoberg and Phillips (2010a, b) and Hoberg et al. (2014). The pairwise correlation coefficients (unreported here) among the four competition measures are all significant at the 1% level.

Of greater interest are the descriptive statistics for the subsamples of R&D and non-R&D firms, presented in Panel B of Table 1. There are 48,177 non-R&D firm-year observations and 61,108 R&D firm-year observations for most of the variables. First, we note that in the last column of the difference-in-means tests, R&D firms are significantly different from non-R&D firms in all of the dimensions. Second, the mean assets of R&D firms are \$383 million greater than the assets of non-R&D firms, whereas the median assets are smaller for R&D firms compared with non-R&D firms, suggesting a larger proportion of small firms in the R&D sample. Third, both the average and median cash of R&D firms are larger than the cash held by non-R&D firms, a \$101 million difference in means. For R&D firms, the average capitalized R&D stock is equivalent to \$315 million with a median of \$20 million.

Turning to some of the other key variables, we see that the average (median) cash-to-assets ratio for the R&D sample is 0.20 (0.11), twice as high as the average (more than double for the median) cash-to-assets ratio for the non-R&D sample. The mean value of R&D-to-assets of R&D firms is 0.08 with a median of 0.04. R&D firms are in the industries that experience higher cash flow volatility and equity issuance volatility, issue more stocks, and appear to be in more competitive industries, than non-R&D firms. R&D firms have lower cash flow and leverage, higher net working capital and market-to-book, lower tangibility, and are less likely to spend on capital investment and acquisitions than non-R&D firms. Using effective tax rate (ETR) as a proxy for uncertain tax positions, we find that firms with R&D have significantly lower ETR than those without R&D, confirming that R&D investment is related with aggressive tax strategies (Hanlon et al., 2016).<sup>7</sup> Finally, we note that the average R&D firm has a significantly lower Altman's (1968) Z-score (implying greater distress risk) than the average non-R&D firm (Acharya et al., 2012).

## 3. Why do R&D firms hold more cash than they used to?—Theory and predictions

It has long been recognized in the academic literature, at least going back to Arrow (1962), that R&D driven innovation is intrinsically difficult to finance using external funds (especially debt) due to its intangible nature, uncertain outcome, and potentially severe asymmetric information problems (e.g., Hall, 2002). Thus, theory suggests that in the cross section, we should find a positive relation between R&D spending and cash holdings. Scholars have found evidence that this is indeed the case (e.g., Opler et al., 1999; Bates et al., 2009).

While several recent studies have attempted to explain the secular increase in cash holdings, few, however, have explicitly explored the role of R&D in explaining the change of cash over time. Bates et al. (2009) present evidence that the typical firm has increased R&D investment, while Falato et al. (2013) show that greater intangible capital (of which R&D is a significant component) limits a firm's debt capacity, thereby reducing financial flexibility. Both studies mainly examine the effects of changing factors on observed increase in cash balances. By contrast, we focus on the changing sensitivity of cash holdings to illuminate how the fundamental relationship between R&D investment and cash holdings has evolved in a time series setting. And we further explore potential explanations for this puzzling evolution.

Bates et al. (2009) argue that the dramatic increase in cash ratio is concentrated in firms with riskier cash flow opportunities, in more recent IPO cohorts, without paying dividends, and experiencing operating losses. In our sample, we find that the average industry cash flow volatility has increased significantly over the last 40 years. But, when we split the sample into R&D and non-R&D firms, we find that cash flow volatility has increased more dramatically among R&D firms than in non-R&D firms (Fig. 2). While there was little difference in the volatility between the two groups (both of which were 0.05) in 1981, cash flow risk in R&D firms had soared to 0.22 in 2004, which was much higher than in non-R&D firms (0.13). Fig. 2 is consistent with other theoretical and empirical work, which shows that the inter-connection between volatility risk and R&D has greatly strengthened over time (e.g., Chun et al., 2004; Comin and Mulani, 2009). For example, Comin and Philippon (2006) document that volatility has increased more in industries that have experienced larger increases in R&D investment. All of these suggest that if cash flow

<sup>6</sup> For the Fitted HHI measure, we have data through 2005; the data for the Text-based HHI is from 1996 to 2007 and for the product market fluidity from 1997 to 2007.

<sup>7</sup> In unreported tests, we also use cash effective tax rate to proxy for tax avoidance behavior. Our major inferences remain the same.



**Table 1**

Summary statistics.

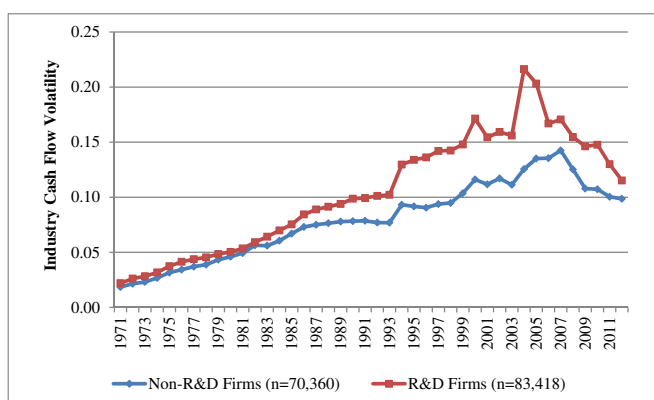
This table reports descriptive statistics for variables of interest for the full sample in Panel A and for subsamples of non-R&D and R&D firms in Panel B. The full sample comprises all publicly traded non-financial and non-utility U.S. industrial firms with positive assets, positive sales, non-negative common equity, and non-missing values for cash and short term equivalents in Compustat from 1980 to 2012, a total of 109,285 firm-year observations. All continuous variables are winsorized at their 1st and 99th percentiles. Dollar values are CPI-adjusted in 2012 dollars. In panel B, R&D firms are defined as firms with non-zero R&D capitalized stock in a given period. The last column shows the difference-in-means tests assuming unequal variances across non-R&D and R&D firms. Variable definitions are reported in [Appendix A](#). \*, \*\*, and \*\*\* indicate statistical significance at the 10%, 5%, and 1% level, respectively.

| Panel A: full sample                         |           |         |         |         |               |         |           |         |                |
|----------------------------------------------|-----------|---------|---------|---------|---------------|---------|-----------|---------|----------------|
| Variable                                     | Mean      | Std.    | Min     | 25th    | Median        | 75th    | Max       |         |                |
| Assets (\$mil)                               | \$1,658   | \$7,932 | \$0     | \$36    | \$150         | \$688   | \$350,574 |         |                |
| Cash (\$mil)                                 | \$142     | \$932   | \$0     | \$2     | \$11          | \$58    | \$73,057  |         |                |
| R&D stock (\$mil, Dep = 15%)                 | \$176     | \$1,385 | \$0     | \$0     | \$1           | \$28    | \$55,968  |         |                |
| Key Variables:                               |           |         |         |         |               |         |           |         |                |
| Cash-to-assets                               | 0.159     | 0.196   | 0.000   | 0.023   | 0.077         | 0.218   | 0.876     |         |                |
| R&D dummy                                    | 0.558     | 0.497   | 0.000   | 0.000   | 1.000         | 1.000   | 1.000     |         |                |
| R&D-to-assets                                | 0.043     | 0.088   | 0.000   | 0.000   | 0.000         | 0.045   | 0.509     |         |                |
| Industry cash flow volatility                | 0.107     | 0.071   | 0.007   | 0.056   | 0.089         | 0.128   | 0.491     |         |                |
| Firm equity issuance volatility              | 0.103     | 0.143   | 0.000   | 0.004   | 0.033         | 0.154   | 0.776     |         |                |
| Industry equity issuance volatility          | 0.108     | 0.054   | 0.000   | 0.065   | 0.104         | 0.149   | 0.712     |         |                |
| Equity issue                                 | 0.050     | 0.147   | 0.000   | 0.000   | 0.003         | 0.017   | 1.098     |         |                |
| Compustat HHI                                | 0.227     | 0.175   | 0.034   | 0.100   | 0.176         | 0.290   | 1.000     |         |                |
| Fitted HHI                                   | 0.058     | 0.022   | 0.032   | 0.044   | 0.052         | 0.066   | 0.255     |         |                |
| Text-based (TNIC) HHI                        | 0.224     | 0.232   | 0.012   | 0.068   | 0.130         | 0.292   | 1.000     |         |                |
| Product market fluidity                      | 6.711     | 3.483   | 0.000   | 4.169   | 6.056         | 8.558   | 27.590    |         |                |
| Import penetration                           | 0.208     | 0.179   | 0.001   | 0.075   | 0.164         | 0.278   | 0.802     |         |                |
| Payout ratio                                 | 0.021     | 0.043   | 0.000   | 0.000   | 0.002         | 0.022   | 0.279     |         |                |
| Size (ln assets)                             | 5.139     | 2.057   | 0.889   | 3.611   | 5.017         | 6.536   | 10.310    |         |                |
| Age                                          | 15.740    | 12.330  | 1.000   | 6.000   | 12.000        | 22.000  | 62.000    |         |                |
| SA-index                                     | −3.077    | 0.860   | −4.638  | −3.653  | −3.096        | −2.506  | −0.661    |         |                |
| WW-index                                     | −0.226    | 0.120   | −0.658  | −0.310  | −0.223        | −0.140  | 0.291     |         |                |
| Controls:                                    |           |         |         |         |               |         |           |         |                |
| CF                                           | 0.020     | 0.198   | −1.011  | 0.008   | 0.070         | 0.115   | 0.305     |         |                |
| NWC                                          | 0.114     | 0.197   | −0.444  | −0.016  | 0.103         | 0.248   | 0.583     |         |                |
| Leverage                                     | 0.225     | 0.190   | 0.000   | 0.051   | 0.201         | 0.352   | 0.979     |         |                |
| MKTBK                                        | 1.850     | 1.578   | 0.175   | 1.018   | 1.337         | 2.001   | 10.930    |         |                |
| Negative income dummy                        | 0.337     | 0.473   | 0.000   | 0.000   | 0.000         | 1.000   | 1.000     |         |                |
| Dividend pay dummy                           | 0.323     | 0.468   | 0.000   | 0.000   | 0.000         | 1.000   | 1.000     |         |                |
| Tangibility                                  | 0.547     | 0.388   | 0.002   | 0.250   | 0.464         | 0.765   | 1.909     |         |                |
| CAPEX                                        | 0.067     | 0.072   | 0.000   | 0.022   | 0.044         | 0.085   | 0.435     |         |                |
| Acquisition cost                             | 0.020     | 0.058   | −0.002  | 0.000   | 0.000         | 0.004   | 0.349     |         |                |
| Debt issue                                   | 0.013     | 0.126   | −0.509  | −0.026  | 0.000         | 0.051   | 0.451     |         |                |
| Effective tax rate (ETR)                     | 0.236     | 0.295   | −1.132  | 0.000   | 0.328         | 0.397   | 1.164     |         |                |
| Rating dummy                                 | 0.538     | 0.499   | 0.000   | 0.000   | 1.000         | 1.000   | 1.000     |         |                |
| Altman's Z-score                             | 1.118     | 3.118   | −14.880 | 0.612   | 1.803         | 2.730   | 6.060     |         |                |
| Panel B: subsamples of R&D and non-R&D firms |           |         |         |         |               |         |           |         |                |
| Variable                                     | R&D firms |         |         |         | Non-R&D firms |         |           |         | Diff.-in-means |
|                                              | Obs       | Mean    | Median  | S.D.    | Obs           | Mean    | Median    | S.D.    |                |
| Assets (\$mil)                               | 61,108    | \$1,827 | \$121   | \$9,263 | 48,177        | \$1,444 | \$192     | \$5,825 | 383***         |
| Cash (\$mil)                                 | 61,108    | \$187   | \$15    | \$1,207 | 48,177        | \$86    | \$8       | \$348   | 101***         |
| R&D stock (\$mil, Dep = 15%)                 | 61,108    | \$315   | \$20    | \$1,842 | 48,177        | \$0     | \$0       | \$0     |                |
| Key Variables:                               |           |         |         |         |               |         |           |         |                |
| Cash-to-assets                               | 61,108    | 0.202   | 0.111   | 0.223   | 48,177        | 0.105   | 0.050     | 0.139   | 0.097***       |
| R&D-to-assets                                | 61,108    | 0.076   | 0.035   | 0.107   | 48,177        | 0.000   | 0.000     | 0.000   |                |
| Industry cash flow volatility                | 61,108    | 0.122   | 0.101   | 0.072   | 48,177        | 0.088   | 0.070     | 0.066   | 0.034***       |
| Firm equity issuance volatility              | 53,327    | 0.114   | 0.032   | 0.155   | 39,017        | 0.088   | 0.033     | 0.123   | 0.026***       |
| Industry equity issuance volatility          | 61,100    | 0.116   | 0.116   | 0.054   | 48,146        | 0.097   | 0.093     | 0.053   | 0.019***       |
| Equity issue                                 | 60,482    | 0.065   | 0.005   | 0.174   | 47,714        | 0.032   | 0.001     | 0.101   | 0.033***       |
| Compustat HHI                                | 61,106    | 0.214   | 0.156   | 0.178   | 48,172        | 0.242   | 0.192     | 0.170   | −0.028***      |
| Fitted HHI                                   | 45,715    | 0.057   | 0.052   | 0.022   | 34,067        | 0.060   | 0.053     | 0.023   | −0.003***      |
| Text-based (TNIC) HHI                        | 23,412    | 0.238   | 0.143   | 0.235   | 17,006        | 0.205   | 0.111     | 0.226   | 0.033***       |
| Product market fluidity                      | 25,610    | 7.024   | 6.411   | 3.656   | 18,353        | 6.279   | 5.639     | 3.179   | 0.745***       |
| Import penetration                           | 27,162    | 0.220   | 0.173   | 0.179   | 5,740         | 0.153   | 0.090     | 0.172   | 0.067***       |
| Payout ratio                                 | 61,108    | 0.022   | 0.001   | 0.045   | 48,177        | 0.019   | 0.003     | 0.041   | 0.002***       |
| Size (ln assets)                             | 61,108    | 5.029   | 4.806   | 2.110   | 48,177        | 5.278   | 5.262     | 1.980   | −0.249***      |
| Age                                          | 61,108    | 17.260  | 13.000  | 13.070  | 48,177        | 13.810  | 10.000    | 11.030  | 3.450***       |
| SA-index                                     | 61,108    | −3.087  | −3.075  | 0.899   | 48,177        | −3.064  | −3.113    | 0.808   | −0.023***      |
| WW-index                                     | 59,774    | −0.218  | −0.211  | 0.126   | 47,555        | −0.235  | −0.236    | 0.112   | 0.017***       |
| Controls:                                    |           |         |         |         |               |         |           |         |                |

(continued on next page)

Table 1 (continued)

| Panel B: subsamples of R&D and non-R&D firms |           |        |        |       |               |       |        |       |                |
|----------------------------------------------|-----------|--------|--------|-------|---------------|-------|--------|-------|----------------|
| Variable                                     | R&D firms |        |        |       | Non-R&D firms |       |        |       | Diff.-in-means |
|                                              | Obs       | Mean   | Median | S.D.  | Obs           | Mean  | Median | S.D.  |                |
| CF                                           | 61,108    | −0.005 | 0.067  | 0.231 | 47,570        | 0.051 | 0.073  | 0.140 | −0.056***      |
| NWC                                          | 61,108    | 0.134  | 0.134  | 0.193 | 48,070        | 0.089 | 0.059  | 0.199 | 0.045***       |
| Leverage                                     | 61,108    | 0.189  | 0.159  | 0.175 | 48,177        | 0.270 | 0.257  | 0.199 | −0.081***      |
| MKTBK                                        | 60,681    | 2.088  | 1.465  | 1.812 | 47,684        | 1.549 | 1.217  | 1.152 | 0.539***       |
| Negative income dummy                        | 61,108    | 0.368  | 0.000  | 0.482 | 48,177        | 0.298 | 0.000  | 0.458 | 0.069***       |
| Dividend pay dummy                           | 61,108    | 0.312  | 0.000  | 0.463 | 48,177        | 0.338 | 0.000  | 0.473 | −0.026***      |
| Tangibility                                  | 61,108    | 0.472  | 0.403  | 0.331 | 48,177        | 0.643 | 0.581  | 0.431 | −0.170***      |
| CAPEX                                        | 60,882    | 0.054  | 0.039  | 0.054 | 48,000        | 0.083 | 0.054  | 0.087 | −0.029***      |
| Acquisition cost                             | 59,401    | 0.019  | 0.000  | 0.055 | 46,951        | 0.022 | 0.000  | 0.061 | −0.003***      |
| Debt issue                                   | 61,047    | 0.007  | 0.000  | 0.119 | 48,161        | 0.020 | 0.000  | 0.135 | −0.013***      |
| Effective tax rate (ETR)                     | 61,098    | 0.212  | 0.295  | 0.304 | 48,174        | 0.267 | 0.358  | 0.280 | −0.055***      |
| Rating dummy                                 | 61,108    | 0.581  | 1.000  | 0.494 | 48,177        | 0.485 | 0.000  | 0.500 | 0.096***       |
| Altman's Z-score                             | 61,108    | 0.683  | 1.749  | 3.574 | 48,177        | 1.667 | 1.881  | 2.307 | −0.984***      |



**Fig. 2.** Industry cash flow volatility from 1971 to 2012 in R&D and Non-R&D firms. This figure shows the yearly averages of industry cash flow volatility for R&D and Non-R&D firms over the period of 1971–2012. R&D firms are defined as firms with non-zero R&D stock in a given period. *Industry Cash Flow Volatility* is the standard deviation of cash flow over assets using 10-year rolling window (if there are at least 3 observations) averaged at the two-digit SIC code level. The sample includes all publicly traded non-financial and non-utility U.S. industrial firms with positive book assets, positive sales, non-negative common equity, and non-missing values for cash and short term equivalents in Compustat from 1971 to 2012, a total of 153,778 firm-year observations. The number of firm-year observations is reported in parentheses.

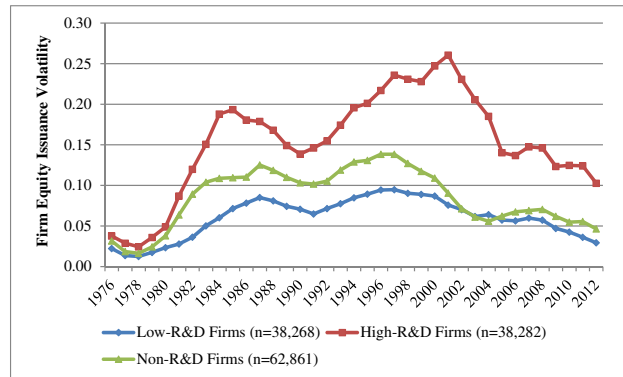
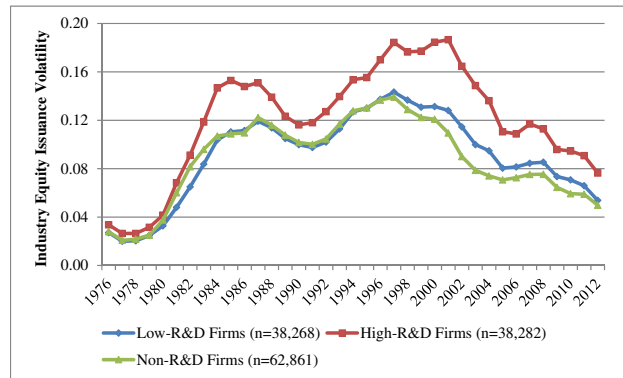
risk is a factor driving the increase in firm cash holdings, the effect would be more pronounced in R&D firms than it would be in non-R&D firms.

Another factor that may have contributed to the divergence of the cash holding behavior of R&D and non-R&D firms is the increased volatility in equity financing. McLean (2011) finds that the propensity to save cash from issuance proceeds has increased over time, which means that the link between equity issuance and cash holdings has strengthened during this period. This is especially pertinent to R&D firms; as Brown et al. (2009) note, R&D investment in manufacturing firms is financed mostly with internal equity (cash flow) and equity issuance. In addition, R&D investment, which often consists of high wage payments to very skilled workers and extensive investments in proprietary information, have high adjustment costs (Brown and Petersen, 2011).

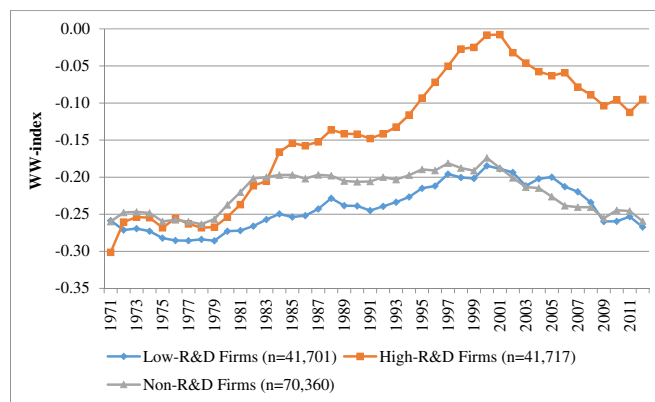
These findings imply that R&D firms would want to hoard cash as a precaution to smooth R&D spending, and this precautionary hoarding will be particularly sensitive to changes in the volatility of equity issuance. Fig. 3 shows that, since 1981, equity issuance volatility has increased far more dramatically at both the firm level (Panel A) and the industry level (Panel B) for high-R&D firms, relative to non-R&D firms.<sup>8</sup> To the extent that firms increasingly issue shares to save cash due to stronger precautionary motives, the effect of equity issuance on cash would be more pronounced in R&D firms than in non-R&D firms.

Prior literature, going as far back as Keynes (1936), suggests that financially constrained firms are more likely to hoard cash. Using the financial constraint index from Whited and Wu (2006), we show in Fig. 4 that, while the average level of financial constraints for non- or low-R&D firms has not changed much in the last three decades, it has increased dramatically for R&D-

<sup>8</sup> High-R&D (or R&D Intensive) firms are defined as firms with R&D-to-sales ratios above the sample medians every year. We obtain qualitatively and quantitatively similar results when we use R&D-to-assets to group firms. To address concern that this phenomenon may be driven by young firms, we exclude firms that went public in the last five years, and we observe similar trends.

A) *Firm-level Equity Issuance Volatility by Non-/Low-/High-R&D Firms*B) *Industry-level Equity Issuance Volatility by Non-/Low-/High-R&D Firms*

**Fig. 3.** Equity issuance volatility from 1976 to 2012 in High, Low, and Non-R&D Firms. This figure shows the yearly average equity issuance volatility for Non-/Low-/High-R&D firms. R&D firms are defined as firms with non-zero R&D stock in a given period. High R&D firms are firms with R&D-to-sales ratios above the sample medium every year. Panel A shows the firm-level *Equity Issuance Volatility*, defined as the standard deviation of equity issues over assets for 5 years rolling window if there are at least 3 observations. Panel B presents the industry-level *Equity Issuance Volatility*, which is the mean firm-level equity issuance volatility at the two-digit SIC code level. The sample includes all publicly traded non-financial and non-utility U.S. industrial firms with positive book assets, positive sales, non-negative common equity, and non-missing values for cash and short term equivalents in Compustat from 1976 to 2012. The number of firm-year observations is reported in parentheses.



**Fig. 4.** Financing constraints from 1971 to 2012 in High, Low, and Non-R&D Firms. This figure shows the yearly average *Whited and Wu (2006)* index for Non-/Low-/High-R&D firms. R&D firms are defined as firms with non-zero R&D stock in a given period. High R&D firms are firms with R&D-to-sales ratios above the sample medium in yearly distribution. WW-index is calculated as  $\{-0.091 * \text{Cash Flow}\} - \{0.062 * \text{Dividend Payer}\} + \{0.021 * \text{Long-Term Debt}\} - \{0.044 * \text{Size}\} + \{0.102 * \text{Industry Growth}\} - \{0.035 * \text{Sales Growth}\}$ , where Long-Term Debt is long term debt over assets, Industry Growth is the four-digit SIC industry sales growth rate and Sales Growth is the firm sales growth rate. The sample includes all publicly traded non-financial and non-utility U.S. industrial firms with positive book assets, positive sales, non-negative common equity, and non-missing values for cash and short term equivalents in Compustat, a total of 153,778 firm-year observations. The number of firm-year observations is reported in parentheses.



intensive firms.<sup>9</sup> Thus, starting from the early 1980s, R&D-intensive firms have operated under increasing financial constraints and would have been expected to hold more cash than other firms in response to this changing operating environment.

Finally, and most importantly, recent research (Della Seta, 2011; Morellec et al., 2013) suggests that product market competition is positively correlated with firm cash holdings. We thus argue that competition is a critical underlying factor in explaining the secular increase in cash for several reasons. Firm cash holdings began a steady rise in the early 1980s, following a wave of deregulation in the 1970s, accompanied by several other reforms across the world that led to an increase in the level of domestic and international competition among U.S. firms. Theory indicates that competition can affect cash holdings directly if firms hoard cash to use as a strategic weapon in competitive industries (Telser, 1966; Frésard, 2010), or as a hedge to ensure corporate survival during downturns. Competition can also increase cash through the impact on other factors. For instance, Irvine and Pontiff (2009) present evidence that, in the four decades leading up to 2003, there were a strong association between competition and stock return idiosyncratic volatility, as well as a strong association between competition and cash flow volatility.

However, even when it comes to competition, there has been a sharp divergence between R&D and non-R&D firms in the past few decades. In Fig. 5, we use two measures of industry concentration – a standard Herfindahl-Hirschman Index (HHI) based on Compustat data at the three-digit SIC industry code (Panel A) and a Fitted HHI (Panel B) developed by Hoberg and Phillips (2010a) – to show the change in industry concentration over this period. We observe that since the early 1980s, the level of competition has increased much more significantly among high-R&D firms than other firms. Indeed, the level of competition among non-R&D firms was not significantly different in 2012 from what it was in 1981. Here again, to the extent that competition may be driving higher levels of cash holdings among U.S. firms, we expect this effect to be especially manifest among R&D firms.

In summary, two clear points emerge from our discussion and review of the current literature. First, as suggested by Bates et al. (2009), Hirschey et al. (2012), and Falato et al. (2013), the amount of resources that innovative firms devote to R&D appears to have increased since 1980. Second, numerous factors that may increase a firm's propensity to hold cash (e.g., cash flow volatility, equity issuance volatility, competition and financial constraints) appears to have increased dramatically in R&D firms over the past three decades, relative to non-R&D firms. The confluence of these two points thus leads us to make three broad predictions that we test in the paper:

**Hypothesis 1.** Due to an increasingly stronger precautionary motive of holding cash among innovating firms relative to others, the growth in cash-to-assets ratio has been more significant in R&D firms than in non-R&D firms between 1980 and 2012.

**Hypothesis 2.** Since 1980, R&D investment has become an increasingly important determinant of cash holdings; and this fundamental shift in the association between R&D and cash explains a significant portion of the increase in recent cash hoards.

**Hypothesis 3.** Changes in the operating and competitive environment play an important role in explaining the increased propensity of R&D firms to hold cash.

#### 4. Cash holdings in R&D and Non-R&D firms from 1980 to 2012

##### 4.1. The cross-sectional determinants of cash holdings between 1980 and 2012

While the focus of our analysis is on understanding the change in the relation between cash and R&D over time, we start by examining the pooled cross-sectional determinants of cash holdings for all firms across our sample period using the empirical model specified below:

$$\text{Cash}_{i,t} = \alpha + \beta_0 * R\&D\_to\_assets_{i,t-1} + \beta_1 * X_{i,t-1} + \epsilon_{i,t} + \gamma_j + d_t \quad (1)$$

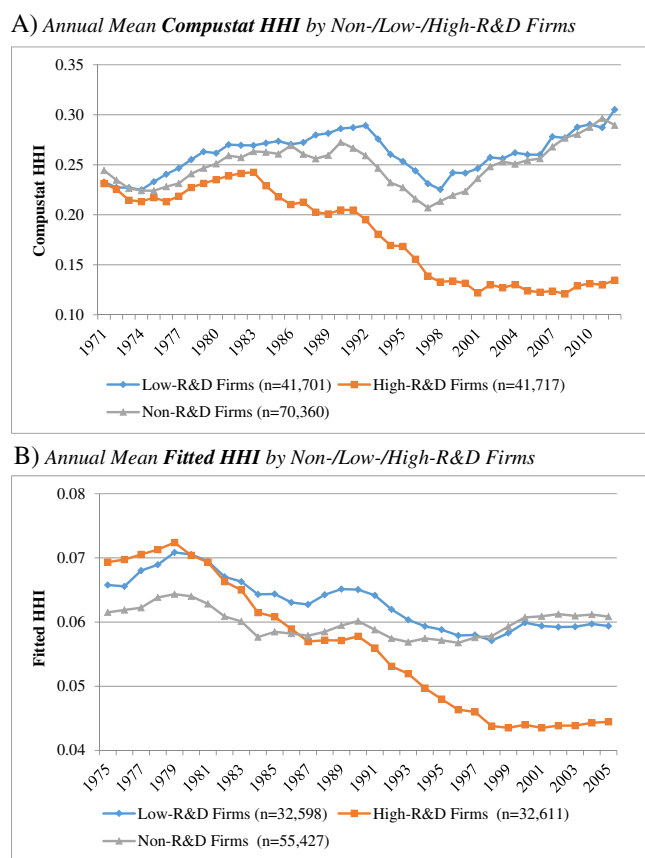
where *Cash* is the firm's cash-to-assets ratio, *R&D-to-assets* is the ratio of R&D expenditures over total assets, *X* is a vector of control variables including *Industry Cash Flow Volatility*, *Equity Issue*, *CF*, *NWC*, *ln(Age)*, *[ln(Age)]<sup>2</sup>*, *Size*, *MKTBK*, *Leverage*, *Negative Income Dummy*, *Dividend Pay Dummy*, *CAPEX*, *Acquisition Cost*, *Debt Issue*, *IPO1 through IPO5*, *Effective Tax Rate*, *Tangibility*, *Rating Dummy*, and *Altman's Z-score*.<sup>10</sup> This set of controls includes the determinants of cash holdings identified in Bates et al. (2009) and other newly discovered factors to mitigate potential omitted variable bias. Eq. (1) includes industry ( $\gamma_j$ ), and year fixed effects ( $d_t$ ).<sup>11</sup>

The results are reported in Table 2. The *t*-statistics are based on robust standard errors clustered by firm (Petersen, 2009). In column (1), we report results from a pooled OLS regression using all firms. To address the issue of changing sample composition, we form a subsample of firms that survived between 1980 and 2012 and report results in column (2). We attempt to control for unobserved, time-invariant factors by incorporating firm fixed effects (in column 3) and using lagged changes (first-differences in column 4).

<sup>9</sup> Whited and Wu index is calculated as  $\{-0.091\text{Cash Flow}\} - \{0.062\text{DividendPayer}\} + \{0.021\text{Long-Term Debt}\} - \{0.044\text{Size}\} + \{0.102\text{Industry Growth}\} - \{0.035\text{Sales Growth}\}$  (Whited and Wu, 2006). Financially constrained firms have high WW index.

<sup>10</sup> For example, *IPO1 through IPO5* are dummies indicating whether a firm had its IPO in the past five years to account for the fact that firms that have recently undergone an initial public offering may hold more cash than other firms (Bouwman and Lowry, 2012).

<sup>11</sup> Industries are defined at the two-digit SIC code level. All results are robust to alternative definitions of industries such as three-digit SIC codes and Fama-French 48 industry classifications.



**Fig. 5.** Industry competition from 1971 to 2012 in High, Low, and Non-R&D firms. This figure shows the changing intensity of competition for Non-/Low-/High-R&D firms. R&D firms are defined as firms with non-zero R&D stock in a given period. High R&D firms are firms with R&D-to-sales ratios above the sample medium in yearly distribution. Compustat HHI is computed as the sum of squared market shares based on firms' sales at three-digit SIC code level. Fitted HHIs are obtained from Hoberg and Phillips library (See Hoberg and Phillips, 2010a). Panel A (B) presents the yearly average Compustat HHI (Fitted HHI) from 1971 (1975) to 2012 (2005). The sample includes all publicly traded non-financial and non-utility U.S. industrial firms with positive book assets, positive sales, non-negative common equity, and non-missing values for cash and short term equivalents in Compustat. The number of firm-year observations is reported in parentheses.

In all specifications, *R&D-to-assets* is a significant determinant of firm cash holdings, both in the full sample and in the more restricted sample of survivors. Among other variables, we find that firms' cash-to-assets ratios are generally positively associated with Altman's Z-score, industry cash flow volatility, equity issuance, cash flow, market-to-book, and debt issuance. Cash-to-assets ratios are negatively related to net working capital, firm size, leverage and propensity to pay dividends. Note, the coefficients of capital investment and acquisition activity are significantly negative, providing evidence that the positive relation between cash and corporate investment is unique to R&D, and do not show up for other forms of investment. The signs and magnitudes of these determinants are similar to those reported in prior literature.

#### A. The evolving effect of R&D on firm cash holdings over time

Next, we turn our analysis to the evolution of cash holdings in R&D and non-R&D firms as described in *Hypothesis (1)*. As we noted in the introduction, Panel A of Fig. 1 illustrates and summarizes the evolution of cash-to-assets ratios for R&D and non-R&D firms from 1971 to 2012. The evidence provides the first piece of evidence in support of our first hypothesis that the increase in cash-to-assets ratio has been more significant in R&D than in non-R&D firms. In Panel B of Fig. 1, we further plot the cash holdings by firms in more or less innovative industries. An industry is defined as innovative if the average R&D-to-assets for the industry during the preceding year is greater than the median average R&D intensity across all industries (Hirshleifer et al., 2012). We see a similar pattern.

In Table 3, we further examine the divergence in the cash-to-assets ratio of R&D and non-R&D firms. The table reports the mean cash-to-assets for each group, and shows that while R&D and non-R&D firms had similar average cash-to-assets ratios of about 10% in 1981, by 2012, the mean cash-to-assets ratio of R&D firms (at 25.35%) was more than twice as high as those of non-R&D firms (at 11.33%). Indeed, the divergence in the cash-to-assets ratios between the two groups widened steadily every year from 1982, peaked at 16.44% in 2003, before leveling out, and falling slightly to 14.02% in 2012.

Univariate analysis does not account for the possibility that the characteristics of R&D firms that are correlated with cash holdings may also have changed between 1980 and 2012, which may explain the change in cash holdings of R&D firms relative to

**Table 2**

Cross-sectional determinants of cash holdings.

This table presents cross-sectional regression results of the determinants of cash holdings between 1980 and 2012. The dependent variable in all regressions is the cash-to-assets ratio. All other variables are as defined in [Appendix A](#). The regression with survivors is limited to firms that survived between 1980 and 2012; all other regressions include all firms in the sample. Year fixed effects are included in all specifications. Industry is defined at the 2-digit SIC code level. The t-statistics in parentheses are based on robust standard errors clustered at the firm level. \*, \*\*, and \*\*\* indicate statistical significance at the 10%, 5%, and 1% level, respectively.

| Dependent variable:<br>cash-to-assets | Baseline (OLS)<br>(1)  | Survivors (80s')<br>(2) | Firm fixed effects<br>(3) | First differences<br>(4) |
|---------------------------------------|------------------------|-------------------------|---------------------------|--------------------------|
| R&D-to-assets                         | 0.573***<br>(31.718)   | 0.357***<br>(4.116)     | 0.069***<br>(3.202)       | 0.053***<br>(3.027)      |
| Effective tax rate                    | −0.005**<br>(−2.396)   | −0.001<br>(−0.211)      | 0.002<br>(1.577)          | 0.001<br>(1.375)         |
| Tangibility                           | −0.089***<br>(−23.343) | −0.077***<br>(−7.043)   | −0.064***<br>(−13.180)    | 0.058***<br>(11.007)     |
| Rating dummy                          | 0.007***<br>(3.104)    | −0.010<br>(−0.334)      | 0.035<br>(0.550)          | 0.000<br>(0.008)         |
| Altman-Z score                        | 0.003***<br>(6.129)    | 0.013***<br>(5.029)     | 0.007***<br>(10.507)      | 0.012***<br>(17.158)     |
| Industry cash flow volatility         | 0.123***<br>(7.564)    | 0.045<br>(1.494)        | 0.052***<br>(3.459)       | −0.006<br>(−0.537)       |
| Equity issue                          | 0.083***<br>(16.738)   | 0.090***<br>(3.366)     | 0.055***<br>(11.475)      | −0.025***<br>(−5.461)    |
| CF                                    | 0.008<br>(1.055)       | −0.049<br>(−1.458)      | −0.032***<br>(−4.546)     | 0.013**<br>(2.236)       |
| NWC                                   | −0.281***<br>(−37.884) | −0.293***<br>(−11.454)  | −0.147***<br>(−21.108)    | 0.077***<br>(13.285)     |
| ln(Age)                               | 0.040***<br>(7.382)    | 0.019<br>(0.644)        | −0.012<br>(−1.464)        | −0.017<br>(−0.884)       |
| [ln(Age)] <sup>2</sup>                | −0.009***<br>(−8.439)  | −0.005<br>(−0.938)      | 0.006**<br>(2.115)        | −0.001<br>(−0.217)       |
| Size                                  | −0.007***<br>(−9.148)  | −0.015***<br>(−6.570)   | −0.020***<br>(−12.418)    | −0.014***<br>(−6.861)    |
| Leverage                              | −0.301***<br>(−48.966) | −0.264***<br>(−13.061)  | −0.143***<br>(−21.717)    | 0.024***<br>(3.707)      |
| MKTBK                                 | 0.006***<br>(9.571)    | 0.010***<br>(2.992)     | 0.004***<br>(6.409)       | 0.001**<br>(1.971)       |
| Negative income dummy                 | 0.001<br>(0.360)       | −0.003<br>(−0.832)      | −0.004***<br>(−3.360)     | 0.001<br>(0.835)         |
| Dividend pay dummy                    | −0.009***<br>(−3.768)  | −0.008<br>(−1.283)      | 0.007***<br>(3.218)       | 0.002<br>(0.921)         |
| CAPEX                                 | −0.287***<br>(−26.565) | −0.377***<br>(−10.522)  | −0.221***<br>(−21.156)    | −0.035***<br>(−4.514)    |
| Acquisition cost                      | −0.342***<br>(−37.014) | −0.277***<br>(−10.556)  | −0.178***<br>(−23.026)    | 0.018***<br>(3.481)      |
| Debt issue                            | 0.090***<br>(19.760)   | 0.106***<br>(7.359)     | 0.033***<br>(7.697)       | −0.003<br>(−0.825)       |
| IPO1                                  | 0.012***<br>(3.020)    | −0.029<br>(−1.419)      | 0.008<br>(1.471)          |                          |
| IPO2                                  | 0.017***<br>(5.215)    | −0.011<br>(−0.728)      | 0.015***<br>(3.664)       | 0.009*<br>(1.825)        |
| IPO3                                  | 0.003<br>(0.937)       | −0.028**<br>(−2.159)    | 0.006*<br>(1.709)         | 0.002<br>(0.550)         |
| IPO4                                  | 0.001<br>(0.432)       | −0.026**<br>(−2.267)    | 0.005*<br>(1.657)         | 0.001<br>(0.471)         |
| IPO5                                  | −0.001<br>(−0.318)     | −0.012<br>(−0.988)      | 0.003<br>(1.460)          | 0.002<br>(0.916)         |
| Firm FE                               | No                     | No                      | Yes                       | No                       |
| Industry FE                           | Yes                    | Yes                     | No                        | Yes                      |
| Year FE                               | Yes                    | Yes                     | Yes                       | Yes                      |
| N                                     | 109,272                | 16,315                  | 109,272                   | 79,189                   |
| Adjusted R <sup>2</sup>               | 0.4668                 | 0.3754                  | 0.7210                    | 0.0487                   |

those of non-R&D firms. Thus, for every year, we run a regression of cash-to-assets ratio on *R&D\_Dummy* and several firm characteristics that may be correlated with both the presence of R&D and the firm's cash holdings using a model taking the form:

$$Cash_{i,t} = \alpha_t + \beta_{0,t} * R\&D\_Dummy_{i,t} + B_{1,t} * X_{i,t} + \epsilon_{i,t} \quad t = 1980, 1981, \dots, 2012 \quad (2)$$

where *Cash* is the firm's cash-to-assets ratio, *R&D\_Dummy* is a variable equal to one if the firm is an R&D firm, and zero otherwise. *X* is the same set of controls as in Eq. (1). Eq. (2) allows us to assess how the effect of being an R&D firm on cash holdings has changed over time in a multivariate setting.

**Table 3**

The increasingly important role of R&D in explaining firm cash holdings from 1980–2012.

This table shows the increasing importance of R&D in driving the secular trend of cash holdings, defined as cash over book assets, from 1980 to 2012. Columns (1) through (4) compare the average cash-to-assets ratios of R&D and non-R&D firms. R&D firms are defined as firms with non-zero R&D capitalized stock in a given period. N represents the number of firm-years. The difference-in-means tests are reported in column (5). The next two columns present the coefficients on R&D\_Dummy (corresponding to Panel A of Fig. 6) and t-statistics in the cross-sectional yearly regression of cash-to-assets ratio on R&D\_Dummy and all control variables, including *Effective Tax Rate*, *Tangibility*, *Rating Dummy*, *Altman's Z-score*, *Industry Cash Flow Volatility*, *Equity Issue*, *CF*, *NWC*, *ln(Age)*,  $[ln(Age)]^2$ , *Size*, *MKTBK*, *Leverage*, *Negative Income Dummy*, *Dividend Pay Dummy*, *CAPEX*, *Acquisition Cost*, *Debt Issue*, *IPO1 through IPO5*. The last two columns report the coefficients on R&D-to-assets (instead of R&D\_Dummy) and the t-statistics, estimated using the same set of controls as indicated above. The t-statistics are based on heteroskedasticity-robust standard errors clustered at the firm level. \*, \*\*, and \*\*\* indicate statistical significance at the 10%, 5%, and 1% level, respectively.

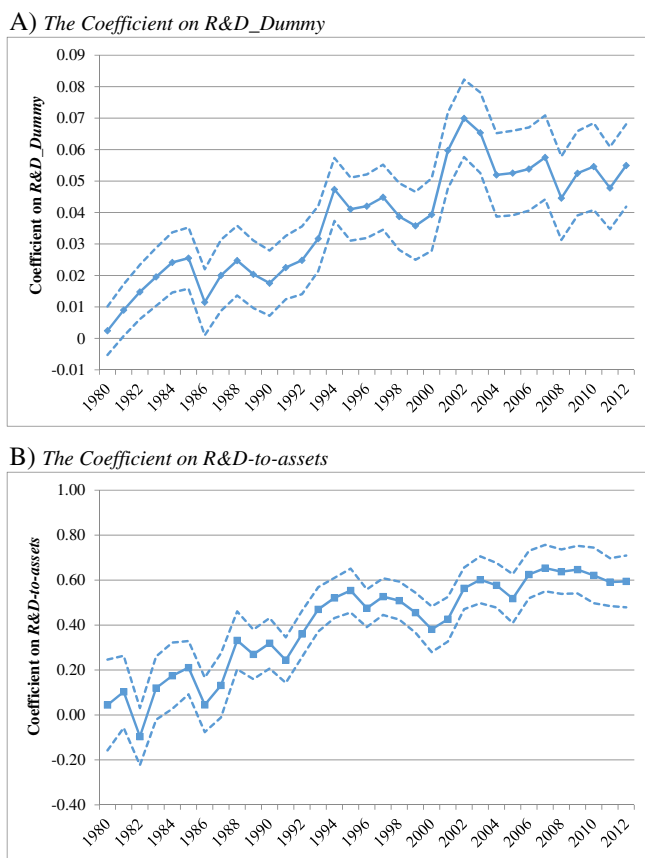
| Year | R&D firms |             | Non-R&D firms |             | Diff.-in-means | R&D_dummy |         | R&D-to-assets |         |
|------|-----------|-------------|---------------|-------------|----------------|-----------|---------|---------------|---------|
|      | N         | Cash/assets | N             | Cash/assets |                | $\beta$   | t-stat. | $\beta$       | t-stat. |
|      | (1)       | (2)         | (3)           | (4)         | (5)            | (6)       | (7)     | (8)           | (9)     |
| 1980 | 1,504     | 0.0831      | 1,315         | 0.0921      | −0.0090**      | 0.0024    | 0.62    | 0.0445        | 0.86    |
| 1981 | 1,467     | 0.0986      | 1,336         | 0.0970      | 0.0016         | 0.0090    | 2.14    | 0.1030        | 1.83    |
| 1982 | 1,671     | 0.1131      | 1,514         | 0.1062      | 0.0070         | 0.0148    | 3.36    | −0.0960       | −2.01   |
| 1983 | 1,730     | 0.1424      | 1,502         | 0.1174      | 0.0250***      | 0.0196    | 4.16    | 0.1197        | 2.32    |
| 1984 | 1,756     | 0.1372      | 1,486         | 0.1067      | 0.0305***      | 0.0241    | 4.96    | 0.1747        | 3.56    |
| 1985 | 1,786     | 0.1441      | 1,517         | 0.1087      | 0.0354***      | 0.0255    | 5.15    | 0.2104        | 5.08    |
| 1986 | 1,761     | 0.1492      | 1,431         | 0.1212      | 0.0280***      | 0.0115    | 2.14    | 0.0451        | 1.01    |
| 1987 | 1,770     | 0.1573      | 1,463         | 0.1204      | 0.0370***      | 0.0200    | 3.48    | 0.1315        | 2.74    |
| 1988 | 1,742     | 0.1548      | 1,485         | 0.1087      | 0.0461***      | 0.0247    | 4.37    | 0.3318        | 7.84    |
| 1989 | 1,702     | 0.1495      | 1,454         | 0.1047      | 0.0448***      | 0.0204    | 3.72    | 0.2696        | 7.03    |
| 1990 | 1,700     | 0.1449      | 1,398         | 0.0979      | 0.0467***      | 0.0176    | 3.32    | 0.3189        | 8.35    |
| 1991 | 1,669     | 0.1574      | 1,384         | 0.1061      | 0.0513***      | 0.0225    | 4.38    | 0.2442        | 6.57    |
| 1992 | 1,747     | 0.1737      | 1,421         | 0.1069      | 0.0668***      | 0.0248    | 4.52    | 0.3605        | 9.64    |
| 1993 | 1,861     | 0.1842      | 1,503         | 0.1042      | 0.0800***      | 0.0317    | 6.00    | 0.4697        | 13.27   |
| 1994 | 1,969     | 0.1831      | 1,678         | 0.0926      | 0.0905***      | 0.0473    | 9.26    | 0.5208        | 16.81   |
| 1995 | 2,010     | 0.1843      | 1,761         | 0.0861      | 0.0982***      | 0.0411    | 8.05    | 0.5535        | 17.89   |
| 1996 | 2,185     | 0.2004      | 1,855         | 0.0939      | 0.1065***      | 0.0420    | 8.16    | 0.4752        | 15.52   |
| 1997 | 2,325     | 0.2183      | 1,937         | 0.0953      | 0.1230***      | 0.0449    | 8.52    | 0.5265        | 17.77   |
| 1998 | 2,240     | 0.2113      | 1,830         | 0.0955      | 0.1158***      | 0.0387    | 7.15    | 0.5088        | 16.92   |
| 1999 | 2,138     | 0.2117      | 1,746         | 0.0888      | 0.1230***      | 0.0358    | 6.50    | 0.4551        | 14.91   |
| 2000 | 2,080     | 0.2282      | 1,688         | 0.0925      | 0.1357***      | 0.0394    | 6.68    | 0.3812        | 10.81   |
| 2001 | 2,035     | 0.2566      | 1,465         | 0.0941      | 0.1626***      | 0.0597    | 9.71    | 0.4262        | 11.82   |
| 2002 | 1,941     | 0.2547      | 1,376         | 0.0958      | 0.1589***      | 0.0699    | 11.14   | 0.5633        | 16.5    |
| 2003 | 1,923     | 0.2698      | 1,343         | 0.1054      | 0.1644***      | 0.0654    | 10.01   | 0.6016        | 16.66   |
| 2004 | 1,894     | 0.2768      | 1,275         | 0.1163      | 0.1605***      | 0.0520    | 7.67    | 0.5768        | 15.61   |
| 2005 | 1,835     | 0.2812      | 1,240         | 0.1244      | 0.1568***      | 0.0526    | 7.67    | 0.5180        | 14.12   |
| 2006 | 1,759     | 0.2731      | 1,225         | 0.1199      | 0.1532***      | 0.0538    | 7.99    | 0.6248        | 16.66   |
| 2007 | 1,737     | 0.2757      | 1,177         | 0.1130      | 0.1627***      | 0.0575    | 8.45    | 0.6535        | 17.16   |
| 2008 | 1,650     | 0.2550      | 1,118         | 0.1051      | 0.1499***      | 0.0446    | 6.58    | 0.6376        | 17.61   |
| 2009 | 1,550     | 0.2626      | 1,088         | 0.1211      | 0.1415***      | 0.0525    | 7.69    | 0.6466        | 16.06   |
| 2010 | 1,499     | 0.2703      | 1,039         | 0.1261      | 0.1442***      | 0.0546    | 7.75    | 0.6211        | 14.95   |
| 2011 | 1,419     | 0.2584      | 1,022         | 0.1210      | 0.1374***      | 0.0478    | 7.17    | 0.5911        | 14.55   |
| 2012 | 1,326     | 0.2535      | 945           | 0.1133      | 0.1402***      | 0.0550    | 8.22    | 0.5942        | 14.32   |

In columns (6) and (7) of Table 3, we report the estimated coefficient on the *R&D\_Dummy* ( $\beta_{0,t}$  from Eq. (2)), for every year between 1980 and 2012, and the *t*-statistic of the estimate, respectively. The results show that, even after controlling for a host of firm characteristics, the cash-to-assets ratio of R&D firms, which was barely higher than that of non-R&D firms in 1980, rose steadily thereafter, and was 5.5% higher by 2012. This result, also illustrated in Fig. 6 (Panel A), demonstrates that the presence of R&D has become a much more important determinant of cash holdings of U.S. firms since 1980.

To further illustrate the changing effect of R&D intensity on cash holdings, in columns (8) and (9) of Table 3, we replace the *R&D\_Dummy* with *R&D-to-assets* ratio in Eq. (2) and report the estimated coefficient (and the corresponding *t*-statistic) for this variable, for every year from 1980 to 2012. The results show that in 1980, after controlling for the set of potential determinants of cash holdings, the average firm held 4.4 cents of cash for every dollar of R&D investment. By 2012, this had risen to 59.4 cents – a fifteen-fold increase. This finding, which is summarized in Fig. 6 (Panel B), provides strong support for Hypothesis (2) that the association of cash holdings and R&D has increased dramatically over the past three decades.

However, it could be that other firm characteristics have had a similarly increasing impact on cash balances as R&D intensity. To examine this possibility, we compare the changing effect of R&D on cash holdings with the changing effects of other factors on cash holdings. We estimate Eq. (1) for successive five-year periods starting with the 1983–1987 period. In Table 4, we report the results of the estimated coefficient on *R&D-to-assets*, as well as the coefficients on *Industry Cash Flow Volatility*, *Dividend Pay Dummy*, *NWC*, and *Equity Issue*.<sup>12</sup> Panel A presents the results for all firms. To account for the possibility that any changes in

<sup>12</sup> We report these selected coefficients because prior literature suggests that changes in these firm characteristics may have affected changes in cash holdings over time – Bates, Kahle, and Stulz (2009) in the cases of *Industry Cash Volatility*, *Dividend Pay Dummy* and *NWC*; Mclean (2011) in the case of *Equity Issue*.



**Fig. 6.** The secular trend in the ability of R&D (*R&D\_Dummy*) and R&D investments (*R&D-to-assets*) to explain cross-sectional differences in cash holdings. The figures below plots the coefficients on *R&D\_Dummy* (Panel A) or *R&D-to-assets* (Panel B) in the cross-sectional yearly regression of cash-to-assets ratio, controlling for *Effective Tax Rate*, *Tangibility*, *Rating Dummy*, *Altman's Z-score*, *Industry Cash Flow Volatility*, *Equity Issue*, *CF*, *NWC*,  $\ln(\text{Age})$ ,  $[\ln(\text{Age})]^2$ , *Size*, *MKTBK*, *Leverage*, *Negative Income Dummy*, *Dividend Pay Dummy*, *CAPEX*, *Acquisition Cost*, *Debt Issue*, *IPO1* through *IPO5*. The sample includes all publicly traded non-financial and non-utility U.S. industrial firms with positive book assets, positive sales, non-negative common equity, and non-missing values for all variables in Compustat from 1980 to 2012. Dash lines indicate the 95% confidence intervals. Standard errors are heteroskedasticity consistent.

the effects of firm factors on cash holdings may be driven by firms that entered the sample after 1980, we again repeat the analysis to the restricted sample of firms that have survived since 1980.

The results in Panel A of Table 4 show that the effect of R&D has increased since early 1980s. The estimated coefficient on *R&D-to-assets* increased from 0.249 ( $t = 5.628$ ) in the 1983–1987 period, to 0.655 ( $t = 15.806$ ) in the 2008–2012 period – a 163% increase. In contrast, there was little change in the coefficients of *Industry Cash Flow Volatility*, *Dividend Pay Dummy*, *NWC*, or *Equity Issue*. We see a similar pattern using a sample restricted to just post-1980 survivors. These results provide compelling evidence that the effect of R&D on cash has increased significantly over time, especially when compared to other important cross-sectional determinants of cash holdings.

In Panel C, we replicate our analysis using the specification from Bates et al. (2009), to provide additional insight into how the effect of R&D has increased over the past several decades. We interact each of our variables with a dummy variable that equals one for all years in the 1990s and after, and another dummy variable for the 2000s. In models (1) and (3), the dependent variable is Cash-to-assets, and the key explanatory variables are *R&D-to-assets* (model 1) and an *R&D\_Dummy* (model 3). The results support those in Tables 3 and Panels A and B of 4. While R&D was a significant determinant of cash holdings relative to assets throughout the entire period, its effect was more significant in the 1990s than in the 1980s and more significant in the 2000s than in the 1990s. We find very similar results in model (2) when cash is measured by *Cash-to-sales* and R&D is by *R&D-to-sales*. In model (4), our dependent variable is Cash-to-assets but we scale R&D by sales; here we find that while the effect of R&D became more significant in the 1990s and after, the difference between the 1990s and 2000s was not significant. Overall, the results in Panel C of Table 4 provide support for our hypothesis that R&D has become a more significant determinant of cash holdings over the past three decades.



**Table 4**

The changing effects of R&D on cash holdings over time.

This table presents OLS regression estimates of selected determinants of cash holdings for six five-year periods starting from 1983. The dependent variable in all regressions is the cash-to-assets ratio. Panel A shows the results using all firms in the sample. Panel B reports the results using firms that survived between 1980 and 2012. The last columns show the differences in coefficients in models (1) and (6) using seemingly unrelated regression system. Panel C presents replications of Bates et al. (2009), Panel B of Table III) by including separate slopes and intercepts for the periods of 1990–2006 and 2000–2006. All other variables are as defined in Appendix A. Industry and year fixed effects are included in all specifications. Intercepts and all other controls are not shown. The t-statistics in parentheses are based on robust standard errors clustered at the firm level. \*, \*\*, and \*\*\* indicate statistical significance at the 10%, 5%, and 1% level, respectively.

| Panel A: all firms                               |                        |                        |                        |                        |                        |                        |                       |                      |                      |                       |                      |                      |
|--------------------------------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|------------------------|-----------------------|----------------------|----------------------|-----------------------|----------------------|----------------------|
| Dependent variable:                              | (1)                    | (2)                    | (3)                    | (4)                    | (5)                    | (6)                    | (6) – (1)             |                      |                      |                       |                      |                      |
| Cash-to-assets                                   | 1983–1987              | 1988–1992              | 1993–1997              | 1998–2002              | 2003–2007              | 2008–2012              | Diff.                 |                      |                      |                       |                      |                      |
| R&D-to-assets                                    | 0.249***<br>(5.628)    | 0.435***<br>(10.467)   | 0.562***<br>(18.150)   | 0.542***<br>(18.690)   | 0.630***<br>(17.841)   | 0.655***<br>(15.806)   | 0.406***<br>(11.557)  |                      |                      |                       |                      |                      |
| Industry cash flow volatility                    | 0.270***<br>(2.626)    | 0.080<br>(0.688)       | 0.073*<br>(1.728)      | –0.147***<br>(–3.006)  | 0.014<br>(0.768)       | –0.069*<br>(–1.771)    | –0.339<br>(–1.192)    |                      |                      |                       |                      |                      |
| Dividend pay dummy                               | –0.006<br>(–1.396)     | –0.013***<br>(–3.009)  | –0.015***<br>(–3.767)  | –0.017***<br>(–3.558)  | –0.012**<br>(–2.509)   | –0.007<br>(–1.457)     | –0.001<br>(–0.557)    |                      |                      |                       |                      |                      |
| NWC                                              | –0.290***<br>(–22.686) | –0.261***<br>(–19.599) | –0.263***<br>(–22.601) | –0.264***<br>(–21.750) | –0.303***<br>(–20.332) | –0.284***<br>(–15.778) | 0.006***<br>(2.874)   |                      |                      |                       |                      |                      |
| Equity issue                                     | 0.045***<br>(3.959)    | 0.061***<br>(5.001)    | 0.115***<br>(13.302)   | 0.081***<br>(8.511)    | 0.083***<br>(6.003)    | 0.072***<br>(4.658)    | 0.027<br>(1.556)      |                      |                      |                       |                      |                      |
| All controls                                     | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    |                       |                      |                      |                       |                      |                      |
| Industry FE                                      | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    |                       |                      |                      |                       |                      |                      |
| Year FE                                          | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    |                       |                      |                      |                       |                      |                      |
| N                                                | 16,605                 | 16,241                 | 19,696                 | 19,109                 | 15,775                 | 12,924                 |                       |                      |                      |                       |                      |                      |
| Adjusted R <sup>2</sup>                          | 0.3221                 | 0.3622                 | 0.4774                 | 0.4942                 | 0.5275                 | 0.5266                 |                       |                      |                      |                       |                      |                      |
| Panel B: survivors only                          |                        |                        |                        |                        |                        |                        |                       |                      |                      |                       |                      |                      |
| R&D-to-assets                                    | 0.107<br>(0.698)       | 0.230<br>(1.596)       | 0.265**<br>(2.205)     | 0.503***<br>(3.254)    | 0.636***<br>(4.565)    | 0.423**<br>(2.432)     | 0.316***<br>(2.900)   |                      |                      |                       |                      |                      |
| Industry cash flow volatility                    | 0.087<br>(0.604)       | 0.069<br>(0.336)       | 0.097<br>(1.033)       | 0.079<br>(1.036)       | 0.011<br>(0.349)       | 0.013<br>(0.232)       | –0.074**<br>(2.364)   |                      |                      |                       |                      |                      |
| Dividend pay dummy                               | 0.001<br>(0.057)       | –0.002<br>(–0.136)     | –0.003<br>(–0.377)     | –0.022**<br>(–2.321)   | –0.014<br>(–1.307)     | –0.009<br>(–0.849)     | –0.010<br>(–1.241)    |                      |                      |                       |                      |                      |
| NWC                                              | –0.317***<br>(–7.571)  | –0.262***<br>(–6.988)  | –0.259***<br>(–6.353)  | –0.249***<br>(–5.813)  | –0.289***<br>(–6.849)  | –0.357***<br>(–9.068)  | –0.040<br>(–0.819)    |                      |                      |                       |                      |                      |
| Equity issue                                     | 0.059<br>(0.944)       | –0.071<br>(–1.229)     | 0.071*<br>(1.714)      | 0.230***<br>(3.367)    | 0.088<br>(1.239)       | 0.187*<br>(1.676)      | 0.128<br>(1.217)      |                      |                      |                       |                      |                      |
| All controls                                     | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    |                       |                      |                      |                       |                      |                      |
| Industry FE                                      | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    |                       |                      |                      |                       |                      |                      |
| Year FE                                          | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    | Yes                    |                       |                      |                      |                       |                      |                      |
| N                                                | 2,483                  | 2,386                  | 2,443                  | 2,462                  | 2,523                  | 2,577                  |                       |                      |                      |                       |                      |                      |
| Adjusted R <sup>2</sup>                          | 0.3457                 | 0.3615                 | 0.3414                 | 0.3809                 | 0.4016                 | 0.4617                 |                       |                      |                      |                       |                      |                      |
| Panel C: using Bates et al. (2009) specification |                        |                        |                        |                        |                        |                        |                       |                      |                      |                       |                      |                      |
| Dependent variable                               | (1)<br>Cash-to-assets  |                        |                        | (2)<br>Cash-to-sales   |                        |                        | (3)<br>Cash-to-assets |                      |                      | (4)<br>Cash-to-assets |                      |                      |
|                                                  | Estimate               | Interaction<br>1990s   | Interaction<br>2000s   | Estimate               | Interaction<br>1990s   | Interaction<br>2000s   | Estimate              | Interaction<br>1990s | Interaction<br>2000s | Estimate              | Interaction<br>1990s | Interaction<br>2000s |
| R&D-to-assets                                    | 0.304***<br>(8.808)    | 0.254***<br>(6.482)    | 0.113***<br>(3.310)    |                        |                        |                        |                       |                      |                      |                       |                      |                      |

(continued on next page)

Table 4 (continued)

| Panel C: using Bates et al. (2009) specification |                        |                       |                       |                        |                       |                      |                        |                       |                       |                        |                       |                       |
|--------------------------------------------------|------------------------|-----------------------|-----------------------|------------------------|-----------------------|----------------------|------------------------|-----------------------|-----------------------|------------------------|-----------------------|-----------------------|
| Dependent variable                               | (1)<br>Cash-to-assets  |                       |                       | (2)<br>Cash-to-sales   |                       |                      | (3)<br>Cash-to-assets  |                       |                       | (4)<br>Cash-to-assets  |                       |                       |
|                                                  | Estimate               | Interaction<br>1990s  | Interaction<br>2000s  | Estimate               | Interaction<br>1990s  | Interaction<br>2000s | Estimate               | Interaction<br>1990s  | Interaction<br>2000s  | Estimate               | Interaction<br>1990s  | Interaction<br>2000s  |
| R&D-to-sales                                     |                        |                       |                       | 1.382***<br>(10.543)   | 0.711***<br>(4.618)   | 0.261**<br>(2.329)   |                        |                       |                       | 0.067***<br>(12.588)   | 0.015**<br>(2.390)    | -0.006<br>(-1.361)    |
| R&D_dummy                                        |                        |                       |                       |                        |                       |                      | 0.020***<br>(6.745)    | 0.016***<br>(4.548)   | 0.028***<br>(6.474)   |                        |                       |                       |
| Industry cash flow                               | 0.179***<br>(3.848)    | 0.069<br>(1.403)      | -0.035<br>(-1.216)    | 1.675***<br>(4.024)    | -1.425***<br>(-3.073) | 0.903***<br>(2.937)  | 0.215***<br>(4.633)    | 0.141***<br>(2.798)   | -0.112***<br>(-3.673) | 0.252***<br>(5.546)    | 0.094*<br>(1.935)     | -0.065**<br>(-2.245)  |
| volatility                                       |                        |                       |                       |                        |                       |                      |                        |                       |                       |                        |                       |                       |
| MKTBK                                            | 0.011***<br>(9.051)    | 0.003*<br>(1.784)     | 0.000<br>(0.032)      | 0.022<br>(1.469)       | -0.006<br>(-0.316)    | 0.021<br>(1.099)     | 0.013***<br>(10.792)   | 0.004***<br>(2.795)   | -0.001<br>(-0.524)    | 0.011***<br>(9.167)    | 0.004**<br>(2.450)    | 0.000<br>(0.175)      |
| Size                                             | -0.010***<br>(-12.103) | 0.005***<br>(5.136)   | 0.006***<br>(4.705)   | -0.044***<br>(-6.941)  | 0.010<br>(1.269)      | 0.004<br>(0.324)     | -0.010***<br>(-11.826) | 0.007***<br>(6.378)   | 0.005***<br>(3.919)   | -0.010***<br>(-11.942) | 0.004***<br>(4.323)   | 0.004***<br>(3.490)   |
| CF                                               | 0.023**<br>(2.034)     | -0.033**<br>(-2.336)  | -0.018<br>(-1.227)    | -0.203<br>(-1.345)     | 0.205<br>(1.007)      | -0.067<br>(-0.314)   | -0.012<br>(-1.072)     | -0.096***<br>(-6.706) | -0.037***<br>(-2.614) | 0.047***<br>(4.423)    | -0.037***<br>(-2.677) | -0.041***<br>(-2.930) |
| NWC                                              | -0.208***<br>(-24.744) | -0.012<br>(-1.191)    | -0.020*<br>(-1.727)   | -1.124***<br>(-12.679) | 0.131<br>(1.265)      | -0.028<br>(-0.230)   | -0.206***<br>(-23.953) | -0.017*<br>(-1.739)   | -0.043***<br>(-3.765) | -0.191***<br>(-22.993) | -0.007<br>(-0.762)    | -0.043***<br>(-3.800) |
| CAPEX                                            | -0.325***<br>(-21.065) | -0.033*<br>(-1.694)   | -0.117***<br>(-4.318) | -0.991***<br>(-5.514)  | 0.071<br>(0.268)      | -0.509<br>(-1.206)   | -0.314***<br>(-20.252) | -0.034*<br>(-1.669)   | -0.097***<br>(-3.533) | -0.321***<br>(-20.915) | -0.038*<br>(-1.956)   | -0.136***<br>(-5.068) |
| Leverage                                         | -0.272***<br>(-31.997) | -0.038***<br>(-3.640) | -0.049***<br>(-3.960) | -0.948***<br>(-10.988) | 0.350***<br>(3.101)   | -0.107<br>(-0.806)   | -0.280***<br>(-33.038) | -0.067***<br>(-6.391) | -0.046***<br>(-3.675) | -0.270***<br>(-32.554) | -0.055***<br>(-5.477) | -0.066***<br>(-5.491) |
| Dividend pay dummy                               | -0.010***<br>(-3.130)  | -0.020***<br>(-5.003) | -0.016***<br>(-3.542) | -0.115***<br>(-6.449)  | 0.018<br>(0.788)      | -0.036<br>(-1.157)   | -0.012***<br>(-3.720)  | -0.027***<br>(-6.926) | -0.017***<br>(-3.766) | -0.013***<br>(-4.029)  | -0.023***<br>(-5.801) | -0.014***<br>(-3.253) |
| Acquisition cost                                 | -0.118***<br>(-8.501)  | -0.096***<br>(-5.585) | -0.058***<br>(-3.073) | -0.384***<br>(-3.552)  | -0.298**<br>(-2.220)  | -0.415**<br>(-2.102) | -0.125***<br>(-8.927)  | -0.094***<br>(-5.296) | -0.080***<br>(-3.957) | -0.118***<br>(-8.564)  | -0.096***<br>(-5.554) | -0.055***<br>(-2.838) |
| N                                                | 93385                  |                       |                       | 93385                  |                       |                      | 93385                  |                       |                       | 93385                  |                       |                       |
| Adjusted R <sup>2</sup>                          | 0.4148                 |                       |                       | 0.4069                 |                       |                      | 0.3864                 |                       |                       | 0.4224                 |                       |                       |

#### 4.2. How much of the change in cash holdings can be attributed to R&D?

Although the effect of R&D on cash holdings has increased since 1980, it still unclear *how much* R&D has contributed to the overall increase in aggregate cash holdings across U.S. firms between 1980 and 2012. We attempt to address this question in this subsection.

As we noted before, there are two channels by which firm characteristics can explain the increase in cash ratios. The first is that the characteristics of the typical firm have changed significantly over time. We term this the *changing characteristics* channel. The second is that the sensitivity of cash holdings to firm characteristics may have itself changed over time, such that even if firm factors had remained the same, the level of cash would still have increased. We term this the *changing sensitivity* channel. In Table 5, we attempt to account for how much each of these channels has contributed to changes in the average cash-to-assets ratio of U.S. firms between 1980 and 2012, and we do this for each determinant of cash holdings.

We first calculate the percentage change in the cash-to-assets ratio arising from the *changing characteristics* channel for each factor as follows. We start by estimating the determinants of cash holdings model from Eq. (1), for all firms for the 1980–1989 period, and note the estimated coefficient on the factor, denoted as  $\beta_{1980-1989}$  and reported in column (3) of Table 5. We then assume that this estimated coefficient persists through the 2003–2012 period. Next, we calculate the change in sample averages for each factor between the 1980–1989 and 2003–2012 periods, as reported in columns (1) and (2). We then obtain the change in cash-to-assets ratio due to that factor by multiplying this change in sample averages by the estimated coefficient from the earlier (1980–1989) period,  $\beta_{1980-1989}$ . Since we are holding the effect of that factor constant and only varying the average firm characteristics, the number that we obtain is the change in cash-to-assets ratio that can be attributed solely to the *changing characteristics* channel. Thus, the change in cash due to the change in a particular firm characteristic,  $X$ , between the 1980s and the 2000s, assuming that the sensitivity of cash to  $X$  ( $\beta_{1980-1989}$ ) had remained constant since the 1980s, is:

$$\Delta Cash_{\text{changing characteristics}} = \beta_{1980-1989} * [X_{2003-2012} - X_{1980-1989}] \quad (3)$$

Column (5) of Table 5 shows the results of this exercise. The average firm had an R&D-to-assets ratio of 2.6% between 1980 and 1989, and this ratio increased by 2.8%, to 5.4% in the 2003–2012 period. If the estimated relationship between R&D and cash-to-assets ratio, which was 0.279 between 1980 and 1989, had remained constant through 2012, the increase in cash-to-assets ratio that was due just to the change in the typical firm's R&D-to-assets ratio would have been  $0.279 * 2.8\% = 0.77\%$ . In other words, if firm's sensitivity to cash holdings had remained constant at the 1980s level, the cash-to-assets ratio of the typical firm would have been 0.77% higher simply based on the increased R&D intensity of the firm.

We move on to calculating the percentage change in the cash-to-assets ratio arising from the *changing sensitivity* channel for each factor as follows. Column (1) of Table 5 shows the sample average for each factor for the 1980 to 1989 period. We assume that this sample persists through the 2003–2012 period. We again estimate Eq. (1) for all firms for both the 1980–1989 and the 2003–2012 periods (these coefficients are reported in columns 3 and 4), and calculate the change in the estimated coefficients for each factor between the two periods. The change in cash-to-assets ratio due to the changing sensitivity of that factor is obtained through multiplying the change in the coefficients by the sample average of the factor from the earlier (1980–1989) period. Since we are holding the characteristic constant and only varying the sensitivity of cash to that factor, the number that we obtain in this case is the change in cash-to-assets ratio that can be attributed solely to the *changing sensitivity* channel. Thus, the change in cash due to the change in the sensitivity of cash to a particular firm characteristic,  $X$ , between the 1980s and the 2000s, assuming that the characteristic itself ( $X_{1980-1989}$ ) had remained constant, is:

$$\Delta Cash_{\text{changing sensitivity}} = [\beta_{2003-2012} - \beta_{1980-1989}] * X_{1980-1989} \quad (4)$$

Column (6) of Table 5 presents the results. Here, the association between R&D and cash, which was 0.279 in the 1980 to 1989 period, increased to 0.657 in the 2003 to 2012 period. If the sample average of R&D-to-assets ratio, which was 2.6% in the 1980s, had remained constant through 2012, the increase in cash-to-assets ratio that was due to just the changing sensitivity of cash to R&D would have been  $(0.657 - 0.279) * 2.6\% = 0.99\%$ . Thus, if the typical firm's R&D-to-assets ratio had remained at the 1980s level, the cash-to-assets ratio of that firm would have been 0.99% higher simply based on the increasing sensitivity of cash holdings to R&D investment, a number that is higher than the 0.77% that is due to changing R&D spending.

A key implication from this exercise is presented in column (7), in which we sum up the change in cash-to-assets ratios due to both the *changing characteristics* and *changing sensitivity* channels. That is, the increasing sensitivity of firm cash holdings to R&D spending, and the fact that R&D investment has increased over time suggest that the average cash-to-assets ratio of U.S. firm would be 1.76% higher in the 2003–2012 period than in the 1980–1989 period. Since the actual increase in the mean cash-to-assets ratio between these periods was 8.53%, it means that R&D itself is able to explain 20.6% of the observed increase in cash over this entire timeframe, consistent with Hypothesis (2). Another way to assess the economic importance of R&D is to note that other than the rapid drop in the NWC held by the average firm (which explains about 30% of the increase in firm cash holdings), R&D investment is the firm characteristic that has been the most important in explaining the increase in cash-to-assets ratios between 1980 and 2012.

## 5. Competition and the cash holdings of R&D firms

The analysis in the previous section presents evidence that the sensitivity of cash to R&D has increased in the past several decades, and that this has had an economically significant impact on firms' cash holdings. From our previous discussion, we posit that changes in the operating and competitive environments faced by R&D-intensive firms may be pushing these firms to hold more cash. In this section, we exhaustively examine the role of competition in the cash holding behavior of R&D and non-R&D firms.

### 5.1. The effect of competition on the cash holdings of R&D and Non-R&D firms

In section III, we highlighted (e.g., in Fig. 5) the fact that, in the past several decades, the level of competition has increased more dramatically in R&D-intensive firms than it has in non-R&D firms. Competition could increase cash holdings directly as firms seek to use cash as a strategic weapon (e.g., Frésard, 2010; Morellec et al., 2013), or indirectly through other factors such as idiosyncratic volatility (Irvine and Pontiff, 2009). Our third hypothesis states that competition will be a more important determinant of cash policies of R&D firms, and we test this conjecture in this sub-section.

We start by examining the effect of domestic competition on cash holdings in R&D and non-R&D firms. We use two proxies to measure the extent of domestic competition. The first proxy is based on the Herfindahl–Hirschman Index (HHI) of industry concentration. We use three different HHI measures: a Compustat HHI measure computed at the three-digit SIC code level, a Fitted HHI, and a Text-based (TNIC) HHI. The second proxy for competition is the product market fluidity, which captures the competitive threats faced by a firm and the changes in rival firms' products relative to the firm. Higher fluidity indicates greater competitiveness. We then estimate Eq. (1) with each competition measure for all firms, and for R&D and non-R&D firms separately. Panel A of Table 6 reports the coefficients on the competition proxies only.

From specifications (1) to (4), we find a strong, significantly positive relation between cash-to-assets ratio and competition for all firms using all four measures, supporting the hypothesis that, in general, firms in more competitive industries hold more cash (as found by Morellec et al. (2013)). When we split the firms, however, we observe a striking pattern. Among R&D firms, cash-to-assets ratios are significantly associated with all of the competition measures; the coefficient estimates are  $-0.053$  ( $t = -6.627$ ),  $-0.352$  ( $t = -4.691$ ),  $-0.099$  ( $t = -11.728$ ) and  $0.010$  ( $t = 13.226$ ) for the Compustat HHI, Fitted HHI, TNIC HHI, and product market fluidity, respectively. In contrast, competition has no effect on the cash holding behavior of non-R&D firms.

We next examine the effect of global competition on the firm cash holdings policies. Our proxy for foreign competition is *Import Penetration*, calculated as the total value of imports divided by the value of shipments minus value of exports plus value of imports, at the four-digit SIC code level for manufacturing industries over 1980–2005. The results, as reported in specification (5), are similar to those for domestic competition measures. While there is a strong, positive association between import penetration and the cash holdings of R&D firms, import penetration has no significant effect on the cash balances of non-R&D firms.

**Table 5**

Decomposition of the increase in cash holdings.

This table decomposes the increase in aggregate cash holdings into changes in firm characteristics (changing factor) and changes in the relation between cash holdings and firm characteristics (changing sensitivity). Columns (1) and (2) report the means of each determinant during 1980–1989 and 2003–2012, respectively. Columns (3) and (4) present the coefficients of each factor for the 1980–1989, and the 2003–2012 periods, respectively. The percentage change in the cash-to-assets ratio arising from the changing factor is reported in column (5), which is calculated as  $[(\beta_{1980-1989} * (X_{2003-2012} - X_{1980-1989}))]$ . The percentage change in the cash-to-assets ratio arising from the changing sensitivity is reported in column (6), which is calculated as  $[(\beta_{2003-2012} - \beta_{1980-1989}) * X_{1980-1989}]$ . The last column sums up the changes reported in columns (5) and (6).

| Determinants                  | $X_{1980-1989}$<br>(1) | $X_{2003-2012}$<br>(2) | $\beta_{1980-1989}$<br>(3) | $\beta_{2003-2012}$<br>(4) | Changing factor<br>(5) | Changing sensitivity<br>(6) | Total<br>(7) |
|-------------------------------|------------------------|------------------------|----------------------------|----------------------------|------------------------|-----------------------------|--------------|
| R&D-to-assets                 | 0.026                  | 0.054                  | 0.279                      | 0.657                      | 0.77%                  | 0.99%                       | 1.76%        |
| Industry cash flow volatility | 0.069                  | 0.143                  | 0.372                      | -0.007                     | 2.79%                  | -2.60%                      | 0.19%        |
| Negative income dummy         | 0.276                  | 0.359                  | -0.009                     | 0.006                      | -0.07%                 | 0.42%                       | 0.35%        |
| Dividend pay dummy            | 0.448                  | 0.274                  | -0.005                     | -0.014                     | 0.09%                  | -0.40%                      | -0.31%       |
| Equity issue                  | 0.035                  | 0.056                  | 0.051                      | 0.073                      | 0.11%                  | 0.08%                       | 0.18%        |
| NWC                           | 0.160                  | 0.064                  | -0.275                     | -0.293                     | 2.66%                  | -0.29%                      | 2.37%        |
| CF                            | 0.034                  | 0.013                  | 0.001                      | 0.022                      | 0.00%                  | 0.07%                       | 0.07%        |
| Size                          | 4.817                  | 5.680                  | -0.011                     | -0.007                     | -0.91%                 | 1.60%                       | 0.69%        |
| Leverage                      | 0.252                  | 0.189                  | -0.257                     | -0.321                     | 1.63%                  | -1.63%                      | 0.01%        |
| MKTBK                         | 1.596                  | 2.012                  | 0.005                      | 0.009                      | 0.23%                  | 0.54%                       | 0.76%        |
| CAPEX                         | 0.083                  | 0.049                  | -0.247                     | -0.308                     | 0.84%                  | -0.51%                      | 0.33%        |
| Acquisition cost              | 0.015                  | 0.024                  | -0.235                     | -0.411                     | -0.22%                 | -0.26%                      | -0.48%       |
| Debt issue                    | 0.019                  | 0.005                  | 0.050                      | 0.125                      | -0.07%                 | 0.14%                       | 0.07%        |
| IPO dummy                     | 0.105                  | 0.032                  | 0.000                      | 0.020                      | 0.00%                  | 0.21%                       | 0.22%        |
| Effective tax rate            | 0.305                  | 0.177                  | 0.000                      | -0.011                     | 0.00%                  | -0.33%                      | -0.33%       |
| Tangibility                   | 0.585                  | 0.504                  | -0.094                     | -0.082                     | 0.76%                  | 0.70%                       | 1.46%        |
| Rating dummy                  | 0.290                  | 0.795                  | 0.009                      | 0.011                      | 0.43%                  | 0.07%                       | 0.50%        |
| Altman-Z score                | 1.859                  | 0.372                  | 0.006                      | 0.002                      | -0.88%                 | -0.73%                      | -1.61%       |
| Total                         |                        |                        |                            |                            |                        |                             | 6.24%        |

To examine the changing effect of competition on the cash holdings of R&D firms, we replicate the model reported in row (1) of Panel A in Table 6 for six sub-periods. The results in Panel B suggest that the effect of competition on cash holdings increased from the 1980s to the early 2000s, before leveling out between 2003 and 2007, and then increased again after 2007. In un-tabulated analysis, we find that this result is not sensitive to different competition proxies, and holds for foreign competition as well. Thus, the effect of competition on the cash holdings of R&D firms has generally increased over time.

## 5.2. The endogeneity of product market competition and firm cash holdings

Empirical evidence suggests that firm financing decisions and industry structure tend to be jointly determined (e.g., Valta, 2012). For instance, cash rich firms are able to use predatory pricing strategies to drive cash poor competitors out of the market (Haushalter et al., 2007), thereby increasing the incentives to hold more cash to ensure corporate survival. It is possible that a similar dynamic may explain the association between competition and the observed cash holding behavior of R&D firms. Large cash reserves enable a firm to challenge rivals' bottom lines by spending more on R&D, thus intensifying competition. This competition may in turn further drive up cash holdings as innovative firms strategically increase precautionary liquidity, leading the cash balances of these firms to increase in an upward spiral. Thus, R&D, competition and cash holdings may be at least in part endogenous, which may explain the association between competition and cash holdings that we observe in R&D firms.

We address this concern by exploiting exogenous variation in industry-level import tariffs as a quasi-natural experiment, following Frésard (2010) and Valta (2012). The international trade literature documents that firms are increasingly exposed to foreign competitors due to the globalization and the openness of trade (e.g., Bernard et al., 2006). Large reductions of import tariff rates, in particular, trigger a sudden increase in the competitiveness by the entry of foreign rivals, and are exogenous to corporate policies, according to a battery of falsification tests by Frésard and Valta (2014).

We take advantage of exogenous tariff rate reductions and adopt a difference-in-differences (DiD) framework to isolate the causal effect of competition on cash holdings for R&D firms. The data on large reductions in import tariff rates in specific industries (at the four-digit SIC level) and years are obtained from Frésard and Valta (2014), who identify 74 affected industries (91 events in total) during 1972–2005. Accordingly, a large tariff cut in an industry-year is defined as a negative change in tariff rate that is three times larger than its average change. These events are not clustered in any specific period, thus alleviating the concern of mix compounding effects that are time-specific (i.e., a single shock).

### 5.2.1. Propensity score matching algorithm

A simple approach to evaluating the effect of tariff rate cuts on cash holdings of R&D firms relative to those of non-R&D firms would be to run an OLS regression of cash holdings on an interaction term of the *R&D\_Dummy* and a variable that captures the events. However, this simple approach has a number of drawbacks. First, as shown, the presence of R&D is significantly associated with increasing cash holdings over our sample period. It is thus possible that R&D firms with higher cash holdings enter the sample after the reductions in tariff rates, leading to a spurious relation between cash holding behavior in R&D firms and increased competition. Second, more generally, firms in industries experiencing tariff reductions could be significantly different from those in unaffected industries; these fundamental but unobservable differences that might be correlated with cash holdings could significantly bias our inference.

To address these identification concerns, we construct a treatment group and a control group using a propensity score matching algorithm for R&D and non-R&D firms separately. Focusing on the first occurrence of tariff cuts in the 74 manufacturing industries, we define treated firms as those operating in industries that experience a tariff cut in a given year. Using this treatment dummy as the dependent variable, we estimate a Logit model to generate the likelihood (i.e., propensity score) that a firm will experience a tariff rate cut using several characteristics including *MKTBK*, *Size*, *CF*, *Leverage*, *Capex*, and *Cash-to-assets*, all measured at the year preceding the event (Frésard and Valta, 2014).

Using the predicted probabilities from the Logit model, we then conduct a propensity score matching with a radius of 1% within each group.<sup>13</sup> Specifically, for each treated R&D (non-R&D) firm, we select matched firms, with replacement, with the propensity scores within  $\pm 1\%$  of each other from the group of R&D (non-R&D) firms that operate in different four-digit SIC code industries during the same year. Our final R&D sample comprises 366 treated firms and 385 matched firms. For non-R&D sample, we find 41 matched firms for 30 treated firms.

To get a sense of how well our matching algorithm works, we compare the firm characteristics of treatment and control firms. Panel A of Table 7 presents tests of the matching variables for the treatment and control groups in the year prior to the tariff reductions. Overall, there are no significant differences in matching observables between the treated and control firms, except that the treated R&D firms have a slightly higher average market-to-book ratio than matched R&D firms ( $p$ -value = 0.069). Hence, the matching process has effectively removed most of the observed differences between the treatment and control groups, and should help to isolate the causal effect of entry threats on firm cash holding responses.

### 5.2.2. Difference-in-differences (DiD) analysis

Panel B of Table 7 shows the average differences in cash holdings from the year before to the year after tariff rate cuts for the treated and matched firms, and the DiD estimates, for R&D and non-R&D firms, respectively. We observe a striking difference in

<sup>13</sup> Results are not sensitive to different cutoffs (e.g., using 5% as the radius for propensity score matching).

**Table 6**

The effect of product market competition on firm cash holdings.

This table presents the effect of competition on firm cash holdings over the 1980–2012 period. All models use cash-to-assets as the dependent variable. Panel A shows OLS regression results for all firms and for subsamples split by R&D. Domestic competition is measured by Compustat HHI, Fitted HHI (Hoberg and Phillips, 2010a), TNIC HHI (Hoberg and Phillips, 2010a, b), Product market fluidity (Hoberg et al., 2014). Foreign competition is proxied by import penetration among manufacturing industries over 1980–2005. Panel B presents the time-varying effect of competition on R&D firm cash holdings. R&D firms are defined as firms with non-zero R&D capitalized stock in a given period. The sample starts with all publicly traded non-financial and non-utility U.S. industrial firms with Compustat coverage from 1980 to 2012. All variables are defined in Appendix A. All the regressions control for year and industry fixed effects (at 2-digit SIC code level). Intercepts and other control variables are included in the regressions but are not shown. The t-statistics in parentheses are based on robust standard errors clustered at the firm level. \*, \*\*, and \*\*\* indicate statistical significance at the 10%, 5%, and 1% level, respectively.

| Panel A: the effect of competition on firm cash holdings since 1980    |                         |                        |                       |                       |                       |                       |                       |                        |        |
|------------------------------------------------------------------------|-------------------------|------------------------|-----------------------|-----------------------|-----------------------|-----------------------|-----------------------|------------------------|--------|
| Dependent variable:                                                    |                         | All firms              |                       |                       | Non-R&D firms         |                       |                       | R&D firms              |        |
| Cash-to-assets                                                         |                         | Estimate               | N                     | Adj. R <sup>2</sup>   | Estimate              | N                     | Adj. R <sup>2</sup>   | Estimate               | N      |
| (1)                                                                    | Compustat HHI           | −0.042***<br>(−7.114)  | 106,384               | 0.4694                | −0.007<br>(−0.761)    | 47,011                | 0.2713                | −0.053***<br>(−6.627)  | 59,373 |
| (2)                                                                    | Fitted HHI              | −0.308***<br>(−5.521)  | 77,590                | 0.4609                | −0.043<br>(−0.503)    | 33,220                | 0.2622                | −0.352***<br>(−4.691)  | 44,370 |
| (3)                                                                    | TNIC HHI                | −0.076***<br>(−11.807) | 39,443                | 0.5586                | −0.012<br>(−1.282)    | 16,720                | 0.3320                | −0.099***<br>(−11.728) | 22,723 |
| (4)                                                                    | Product market fluidity | 0.008***<br>(14.146)   | 42,940                | 0.5645                | 0.001<br>(1.022)      | 18,057                | 0.3389                | 0.010***<br>(13.226)   | 24,883 |
| (5)                                                                    | Import penetration      | 0.038***<br>(3.044)    | 32,898                | 0.5474                | 0.038<br>(1.125)      | 5,740                 | 0.2927                | 0.036***<br>(2.641)    | 27,158 |
| Panel B: the effect of competition on R&D firm cash holdings over time |                         |                        |                       |                       |                       |                       |                       |                        |        |
| Dependent variable:                                                    |                         | R&D firms              |                       |                       |                       |                       |                       |                        |        |
| Cash-to-assets                                                         |                         | 1983–1987<br>(1)       | 1988–1992<br>(2)      | 1993–1997<br>(3)      | 1998–2002<br>(4)      | 2003–2007<br>(5)      | 2008–2012<br>(6)      | Diff.<br>(6) − (1)     |        |
| Compustat HHI                                                          |                         | −0.022<br>(−1.558)     | −0.045***<br>(−3.210) | −0.081***<br>(−5.764) | −0.089***<br>(−5.282) | −0.052***<br>(−3.173) | −0.080***<br>(−4.522) | −0.058*<br>(−1.88)     |        |
| All controls                                                           |                         | Yes                    | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   |                        |        |
| Industry FE                                                            |                         | Yes                    | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   |                        |        |
| Year FE                                                                |                         | Yes                    | Yes                   | Yes                   | Yes                   | Yes                   | Yes                   |                        |        |
| N                                                                      |                         | 8,992                  | 8,832                 | 10,730                | 10,834                | 9,399                 | 7,618                 |                        |        |
| Adjusted R <sup>2</sup>                                                |                         | 0.3715                 | 0.4340                | 0.5312                | 0.5161                | 0.5224                | 0.5238                |                        |        |

DiD estimates between R&D and non-R&D firms. Treated R&D firms significantly increase their cash reserves by 3.7% ( $t = 2.322$ ) relative to matched R&D firms over a two-year period surrounding the competitive shock of a tariff rate reduction, while treated non-R&D firms have a negligible (statistically insignificant) increase of 0.13% compared with matched counterparts.

We next turn to multivariate regression analysis by estimating a difference-in-differences regression of the model taking the form:

$$Cash_{i,t} = \alpha + \beta_0 * Treat_i * Post\_cut_{i,t} + \beta_1 * Treat_i + \beta_2 * Post\_cut_{i,t} + \beta_3 * Y_{i,t-1} + \epsilon_{i,t} + \gamma_j + d_t \quad (5)$$

where *Cash* is the cash-to-assets ratio, *Treat* is an indicator equal to one if a firm operates in an industry  $j$  that is affected by a tariff rate reduction, and zero otherwise. *Post\_cut* is a dummy equal to one if an industry  $j$  has undergone a tariff rate cut by the time  $t$ , and zero otherwise.  $Y$  is a vector of controls including *R&D-to-assets* and those specified in Eq. (1). We control for industry ( $\gamma_j$ ) and year fixed effects ( $d_t$ ), and cluster standard errors by firm. The model is estimated on R&D and non-R&D samples separately.

The DiD estimate,  $\beta_0$ , is the coefficient on the interaction term of *Treat* and *Post\_cut*, which captures the causal effect of foreign competition on firm cash holdings. If competition causes R&D firms to accumulate more cash than non-R&D firms, one should observe a positive and significant coefficient on the DiD estimate for R&D firms, but not for non-R&D firms.

The first four columns of Panel C in Table 7 display the DiD estimates for R&D firms and the next four columns for non-R&D firms. In columns (1) and (5), we focus on a window of one year before and one year after the event ( $[-1, +1]$ ); the coefficients on the interaction term have values of 0.028 ( $t = 3.022$ ) for R&D firms, 0.016 ( $t = 0.594$ ) for non-R&D firms, suggesting that treated R&D firms significantly increased their cash holdings by 2.8% more than unaffected R&D firms following a shock to firms' competitive environment. There is no such increase in non-R&D firms. In columns (2)–(3) (columns 6–7), we extend the event windows to  $[-1, +2]$ , and  $[-2, +2]$ , respectively. The positive and significant coefficients on the DiD estimates, observed only in the sample of R&D firms, confirm that R&D firms had a strong cash holding reaction to intensified competition, while non-R&D firms did not.

Finally, in columns (4) and (8), we perform a falsification test that uses year ( $t-3$ ) relative to the actual event year ( $t$ ) and examine the cash holding behavior for R&D and non-R&D firms during  $[-4, -2]$  event window. We do not obtain significant



coefficients on the DiD estimates, supporting that the observed increase in cash holdings among R&D firms is likely caused by the tariff rate changes.

### 5.3. The interaction between competition and financing constraints in explaining firm cash holdings

As the impact of competition on cash holdings may be exacerbated by financial constraints, we examine how constraints and competition interact with the cash holdings of R&D firms relative to those of non-R&D firms. To do this, we divide our R&D and non-R&D samples into “Constrained” and “Unconstrained” groups based on the median values of financing constraint measures in previous year. Our proxies for financial constraints include payout ratio (e.g., Fazzari et al., 1988; Almeida and Campello, 2010), size (e.g., Gertler and Gilchrist, 1994; Almeida et al., 2004), age (e.g., Brown and Petersen, 2011), the “size and age” SA-index (Hadlock and Pierce, 2010), and the WW-index (Whited and Wu, 2006). We note that all the pairwise correlations among these measures of financial constraints are significant at the 1% level, and the correlation coefficients range from 0.10 to 0.88.

In Table 8, we present the results of multivariate regressions (similar to that specified in Eq. (1)) for each of the four groups we described above: Non-R&D/Unconstrained, Non-R&D/Constrained, R&D/Unconstrained and R&D/Constrained. For brevity, only the coefficients on the competition measures are reported for each of our financial constraint proxies, and we report results using Compustat HHI (Panel A), the Fitted HHI (Panel B), TNIC HHI (Panel C), and Product Market Fluidity (Panel D).

The results show a very different pattern between R&D and non-R&D firms. For non-R&D firms, we find that competition has almost no effect on the cash holdings of either constrained or unconstrained firms. Hence, we conclude that, in general, non-R&D firms do not increase their cash holdings in response to increased competition even when they are financially constrained. In contrast, competition has a significant effect for R&D firms, whether or not they are constrained. Further, the effect of competition on cash holdings is stronger for constrained R&D firms. Thus, while competition and financial constraints are important determinants of cash holdings, their significance is only salient in the presence of R&D.

## 6. Conclusion

This paper examines the role of R&D investment in the secular increase in the cash holdings of U.S. firms over the past three decades. Using a large sample of publicly traded U.S. industrial firms with Compustat coverage from 1980 to 2012, this study finds strong evidence that the acceleration of the mean cash-to-assets ratio was almost entirely driven by R&D intensive firms. The positive association between R&D investment and cash holdings was fifteen times more significant in 2012 than in 1980. Moreover, R&D itself explains more than 20% of the increase in aggregate cash holdings over this period.

To understand why the cash holdings of R&D firms have increased so dramatically, we posit several explanations. We show that the operating, financing, and competitive landscape has become more challenging for R&D intensive firms. The growing cash accumulation of R&D firms may explain why the aggregate R&D spending is so smooth even though the key sources of finance have become increasingly volatile (Brown and Petersen (2011)). We find that intensified domestic and international competition among R&D firms magnifies the marginal impact of financial constraints upon the decisions to hold cash.

The combination of these elements makes strategic cash reserves more valuable for R&D intensive firms than for other firms. While a systematic study of the value of cash among R&D firms is outside the scope of this paper, we note that Bates et al. (2011) do find that the marginal value of cash increased significantly more for high R&D firms than for other firms over 1989–2009, a finding that is in line with the results in this paper.

Overall, the findings in this study contribute to the understanding of the role of R&D in determining not only the cross-sectional variation of firm cash holdings policies, but also the change of cash over time. Our work complements recent explanations of the cash holdings puzzle by highlighting a key mechanism that accounts for this secular trend—the evolution in the financing of innovative firms when confronted with increasing global and domestic competition.

## Appendix A. variable definitions.

All names in parentheses refer to the Compustat item names.

*Size*: Natural logarithm of book assets (AT) that is converted to 2012 dollars using the CPI from the US Bureau of Labor Statistics.

*Age*: the number of years the firm is listed with a non-missing stock price on Compustat

*Cash-to-assets*: Cash plus marketable securities (CHE), divided by book value of total assets (AT)

*CF (Cash Flow/TA)*: Cash flow is defined as operating income before depreciation (OIBDP), less interest (XINT) and taxes (TXT), scaled by book assets (AT).

*Market-to-Book (MKTBK)*: Total assets (AT) minus book value of common equity (CEQ) plus the market value of common equity (PRCC\_F \* CSHO), divided by book value of assets (AT)

*NWC (Net Working Capital/TA)*: The ratio of working capital (ACT – LCT) minus cash and marketable securities (CHE) to total assets (AT).

*Leverage*: The ratio of long-term debt (DLTT) plus debt in current liabilities (DLC) to total assets (AT).

*R&D Stock*: Capitalized past R&D expenditures (XRD) calculated using the perpetual inventory method, with 15% annual depreciation rate (See Hall et al. (2005)).

*R&D Firms*: Firms with non-zero R&D capitalized stock in a given period.

**Table 7**

The impact of competition on R&D firm cash holdings using reductions of import tariff rates – difference-in-differences tests.

This table presents diagnostics and results of the difference-in-differences tests on the effect of competition on cash holdings using tariff rate reduction as a quasi-natural experiment. Sample selection starts with all publicly traded manufacturing firms with non-missing matching variables and trade data from 1972 to 2005. For each treated R&D (non-R&D) firm that has experienced tariff rate reduction, we select matched firms using propensity score matching (with replacement and a radius of 1%) from the group of R&D (non-R&D) firms that operate in different four-digit SIC code industries during the same year. Firms are matched in the year prior to the tariff rate cuts by *MKTBK*, *Size*, *CF*, *Leverage*, *Capex*, and *Cash-to-assets*. Panel A presents the results of balance tests of matched treatment and control groups, for R&D and non-R&D firms, respectively. Panel B shows the differences in mean cash holdings from the year before to the year after tariff rate cuts for treated R&D (non-R&D) firms (average treated differences), the matched R&D (non-R&D) firms (average matched differences), and the differences between treated and matched R&D (non-R&D) firms (difference-in-differences). Panel C shows the multivariate analysis of cash holdings around reductions in import tariffs for R&D and non-R&D firms. All variables are defined in [Appendix A](#). Intercepts are included in the regressions but not shown. The t-statistics in parentheses are based on robust standard errors clustered by firm. \*, \*\*, and \*\*\* indicate statistical significance at the 10%, 5%, and 1% level, respectively.

| Panel A: balance tests |                     |                   |        |         |                    |                  |        |         |  |
|------------------------|---------------------|-------------------|--------|---------|--------------------|------------------|--------|---------|--|
|                        | R&D firms           |                   |        |         | Non-R&D firms      |                  |        |         |  |
|                        | treatment (n = 366) | Control (n = 385) | Diff.  | p-Value | Treatment (n = 30) | Control (n = 41) | Diff   | p-Value |  |
| MKTBK                  | 1.662               | 1.505             | 0.157* | 0.069   | 1.253              | 0.972            | 0.282  | 0.118   |  |
| Size                   | 4.86                | 4.90              | −0.036 | 0.775   | 4.527              | 4.617            | −0.091 | 0.715   |  |
| CF                     | 0.02                | 0.03              | −0.006 | 0.687   | 0.057              | 0.072            | −0.015 | 0.433   |  |
| Leverage               | 0.214               | 0.227             | −0.014 | 0.246   | 0.266              | 0.315            | −0.048 | 0.222   |  |
| CAPEX                  | 0.063               | 0.061             | 0.002  | 0.668   | 0.059              | 0.056            | 0.003  | 0.800   |  |
| Cash-to-assets         | 0.128               | 0.128             | 0.000  | 0.986   | 0.074              | 0.073            | 0.001  | 0.946   |  |

| Panel B: univariate tests of mean changes of cash holdings during [−1,+1] of tariff rate reductions |                             |                           |                     |                             |                           |                  |
|-----------------------------------------------------------------------------------------------------|-----------------------------|---------------------------|---------------------|-----------------------------|---------------------------|------------------|
|                                                                                                     | R&D firms                   |                           |                     | Non-R&D firms               |                           |                  |
|                                                                                                     | treatment<br>(After-before) | Control<br>(After-before) | DiD                 | Treatment<br>(After-before) | Control<br>(After-before) | DiD              |
| Cash-to-assets                                                                                      | 0.012<br>(1.169)            | −0.025**<br>(−2.037)      | 0.037***<br>(2.322) | 0.007<br>(0.270)            | −0.007<br>(−0.305)        | 0.013<br>(0.405) |

| Panel C: multivariate difference-in-differences tests |                     |                     |                     |                               |                    |                    |                    |                               |
|-------------------------------------------------------|---------------------|---------------------|---------------------|-------------------------------|--------------------|--------------------|--------------------|-------------------------------|
| Dependent variable:<br>Cash-to-assets                 | R&D Firms           |                     |                     |                               | Non-R&D firms      |                    |                    |                               |
|                                                       | [−1,+1]<br>(1)      | [−1,+2]<br>(2)      | [−2,+2]<br>(3)      | Placebo(−3)<br>[−4,−2]<br>(4) | [−1,+1]<br>(5)     | [−1,+2]<br>(6)     | [−2,+2]<br>(7)     | Placebo(−3)<br>[−4,−2]<br>(8) |
| Treat * Post_cut                                      | 0.028***<br>(3.022) | 0.022***<br>(2.666) | 0.018**<br>(2.206)  | −0.008<br>(−0.846)            | 0.016<br>(0.594)   | 0.014<br>(0.670)   | −0.005<br>(−0.326) | −0.008<br>(−0.321)            |
| Treat                                                 | −0.013<br>(−1.395)  | −0.012<br>(−1.335)  | −0.008<br>(−0.970)  | 0.004<br>(0.390)              | −0.023<br>(−0.959) | −0.017<br>(−0.757) | 0.000<br>(0.015)   | 0.033<br>(1.075)              |
| Post_cut                                              | −0.015*<br>(−1.884) | −0.013*<br>(−1.753) | −0.013*<br>(−1.866) | 0.004<br>(0.497)              | −0.016<br>(−0.755) | −0.012<br>(−0.636) | 0.003<br>(0.155)   | 0.015<br>(0.526)              |
| Controls                                              | Yes                 | Yes                 | Yes                 | Yes                           | Yes                | Yes                | Yes                | Yes                           |
| Industry FE                                           | Yes                 | Yes                 | Yes                 | Yes                           | Yes                | Yes                | Yes                | Yes                           |
| Year FE                                               | Yes                 | Yes                 | Yes                 | Yes                           | Yes                | Yes                | Yes                | Yes                           |
| N                                                     | 1,960               | 2,458               | 3,078               | 1,604                         | 169                | 208                | 269                | 147                           |
| Adjusted R <sup>2</sup>                               | 0.4213              | 0.4219              | 0.4291              | 0.4124                        | 0.3265             | 0.3936             | 0.3791             | 0.6037                        |

*R&D Dummy*: An indicator variable equals one for “R&D Firms”, and zero otherwise.

*R&D-to-assets*: The ratio of R&D expenditure (XRD) to assets (AT); missing values are set to zero.

*Equity Issue*: The sale of common and preferred stock (SSTK), divided by total assets (AT).

*Firm Equity Issuance Volatility*: The standard deviation of equity issues (SSTK) over assets (AT) using 5 years rolling window if there are at least 3 observations for the firm.

*Effective Tax Rate (ETR)*: The ratio of tax expense (TXT) to pre-tax income (PI) (Lisowsky et al., 2013)

*Tangibility*: The ratio of gross property, plant, and equipment (PPEGT) to total assets (AT).

*Rating Dummy*: An indicator equals one if a firm has an S&P credit rating, and zero otherwise.

*Altman's Z-Score*: computed as  $\{3.3 * \text{pre-tax income (PI)}/\text{total assets (AT)}\} + \text{sales (SALE)} + \{1.4 * \text{retained earnings (RE)}/\text{total assets (AT)}\} + \{1.2 * [\text{current assets (ACT)} - \text{current liabilities (LCT)}/\text{total assets (AT)}]\}$  (Altman, 1968; Lemmon and Roberts, 2010)

*Dividend Pay Dummy*: An indicator equals one if a common dividend (DVC) is positive, zero otherwise.

*CAPEX*: The ratio of Capital expenditures (CAPX) to total assets (AT).

*Acquisition Cost*: The ratio of Acquisitions (AQC) to total assets (AT).

*Negative Income Dummy*: An indicator equals one if a firm suffers operating loss (NI) in a given year.

*SA index*: The index is based on Hadlock and Pierce (2010) and calculated as  $\{-0.737 * \text{Size}\} + \{0.043 * \text{Size}^2\} - \{0.040 * \text{Age}\}$ , Size equals the log of inflation-adjusted book assets, and Age is the number of years the firm is listed with a

**Table 8**

The interaction between competition and financing constraints in explaining firm cash holdings.

This table presents OLS regression results of the effect of competition on cash holdings of R&D and non-R&D firms in constrained/unconstrained status. All models use cash-to-assets as the dependent variable. Panels A-D show the coefficient estimates of Compustat HHI, Fitted HHI, TNIC HHI, and Product market fluidity, respectively. Firms are classified into Constrained/Unconstrained groups based on the median values of financing constraint proxies in the previous year. The sample starts with all publicly traded non-financial and non-utility U.S. industrial firms with Compustat coverage from 1980 to 2012. All variables are defined in Appendix A. All the regressions incorporate the full set of controls, and year and industry fixed effects (at 2-digit SIC code level). The t-statistics in parentheses are based on robust standard errors clustered at the firm level. \*, \*\*, and \*\*\* indicate statistical significance at the 10%, 5%, and 1% level, respectively.

| Panel A: competition measured by compustat HHI           |                         | Non-R&D firms        |                    | R&D firms             |                       |
|----------------------------------------------------------|-------------------------|----------------------|--------------------|-----------------------|-----------------------|
| Financing constraints criteria                           |                         | Unconstrained<br>(1) | Constrained<br>(2) | Unconstrained<br>(3)  | Constrained<br>(4)    |
| Payout Ratio                                             | Compustat HHI           | −0.011<br>(−0.943)   | −0.003<br>(−0.258) | −0.045***<br>(−4.761) | −0.054***<br>(−4.684) |
| Size                                                     | N                       | 25,324               | 22,845             | 29,895                | 31,201                |
|                                                          | Compustat HHI           | −0.007<br>(−0.663)   | 0.002<br>(0.116)   | −0.036***<br>(−4.041) | −0.046***<br>(−3.780) |
| Age                                                      | N                       | 26,795               | 21,374             | 28,374                | 32,722                |
|                                                          | Compustat HHI           | −0.003<br>(−0.252)   | −0.009<br>(−0.853) | −0.034***<br>(−3.781) | −0.068***<br>(−5.202) |
| SA-index                                                 | N                       | 25,741               | 22,428             | 34,671                | 26,425                |
|                                                          | Compustat HHI           | −0.010<br>(−1.001)   | 0.003<br>(0.192)   | −0.032***<br>(−3.545) | −0.056***<br>(−4.314) |
| WW-index                                                 | N                       | 26,229               | 21,940             | 30,821                | 30,275                |
|                                                          | Compustat HHI           | −0.008<br>(−0.804)   | 0.004<br>(0.326)   | −0.034***<br>(−3.802) | −0.048***<br>(−3.948) |
|                                                          | N                       | 24,728               | 23,441             | 26,445                | 34,651                |
| Panel B: competition measured by fitted HHI              |                         |                      |                    |                       |                       |
| Payout ratio                                             | Fitted HHI              | 0.025<br>(0.227)     | −0.098<br>(−0.980) | −0.203***<br>(−2.643) | −0.457***<br>(−4.140) |
| Size                                                     | N                       | 17,599               | 15,621             | 21,890                | 22,480                |
|                                                          | Fitted HHI              | −0.002<br>(−0.018)   | −0.030<br>(−0.206) | −0.151**<br>(−2.011)  | −0.384***<br>(−2.981) |
| Age                                                      | N                       | 18,689               | 14,531             | 21,149                | 23,221                |
|                                                          | Fitted HHI              | 0.094<br>(0.836)     | −0.131<br>(−1.146) | −0.147*<br>(−1.959)   | −0.565***<br>(−4.074) |
| SA-index                                                 | N                       | 18,226               | 14,994             | 25,551                | 18,819                |
|                                                          | Fitted HHI              | 0.018<br>(0.192)     | −0.065<br>(−0.446) | −0.141*<br>(−1.884)   | −0.450***<br>(−3.360) |
| WW-index                                                 | N                       | 18,438               | 14,782             | 22,770                | 21,600                |
|                                                          | Fitted HHI              | 0.033<br>(0.355)     | −0.074<br>(−0.541) | −0.145**<br>(−2.004)  | −0.415***<br>(−3.346) |
|                                                          | N                       | 17,274               | 15,946             | 19,680                | 24,690                |
| Panel C: competition measured by TNIC HHI                |                         |                      |                    |                       |                       |
| Payout ratio                                             | TNIC HHI                | −0.007<br>(−0.578)   | −0.012<br>(−1.064) | −0.081***<br>(−8.191) | −0.104***<br>(−8.583) |
| Size                                                     | N                       | 9,337                | 7,383              | 11,291                | 11,432                |
|                                                          | TNIC HHI                | −0.016<br>(−1.582)   | −0.005<br>(−0.381) | −0.091***<br>(−8.746) | −0.086***<br>(−7.738) |
| Age                                                      | N                       | 10,596               | 6,124              | 10,709                | 12,014                |
|                                                          | TNIC HHI                | 0.003<br>(0.248)     | −0.017<br>(−1.366) | −0.075***<br>(−7.915) | −0.118***<br>(−7.404) |
| SA-index                                                 | N                       | 9,217                | 7,503              | 13,007                | 9,716                 |
|                                                          | TNIC HHI                | −0.015<br>(−1.471)   | −0.002<br>(−0.177) | −0.080***<br>(−8.319) | −0.096***<br>(−7.248) |
| WW-index                                                 | N                       | 10,421               | 6,299              | 11,810                | 10,913                |
|                                                          | TNIC HHI                | −0.012<br>(−1.105)   | −0.003<br>(−0.245) | −0.083***<br>(−8.844) | −0.094***<br>(−8.284) |
|                                                          | N                       | 9,821                | 6,899              | 9,930                 | 12,793                |
| Panel D: competition measured by product market fluidity |                         |                      |                    |                       |                       |
| Payout ratio                                             | Product market fluidity | 0.001<br>(1.238)     | 0.001<br>(0.689)   | 0.009***<br>(9.007)   | 0.010***<br>(9.865)   |
| Size                                                     | N                       | 10,106               | 7,951              | 12,336                | 12,547                |
|                                                          | Product market fluidity | 0.001**<br>(2.166)   | −0.001<br>(−0.671) | 0.007***<br>(8.119)   | 0.010***<br>(8.852)   |
| Age                                                      | N                       | 11,471               | 6,586              | 11,629                | 13,254                |
|                                                          | Product market fluidity | 0.000<br>(0.319)     | 0.001<br>(0.997)   | 0.009***<br>(7.742)   | 0.011***<br>(10.261)  |

(continued on next page)

Table 8 (continued)

| Panel A: competition measured by compustat HHI |                         | Non-R&D firms        |                    | R&D firms            |                     |
|------------------------------------------------|-------------------------|----------------------|--------------------|----------------------|---------------------|
| Financing constraints criteria                 |                         | Unconstrained<br>(1) | Constrained<br>(2) | Unconstrained<br>(3) | Constrained<br>(4)  |
| SA-index                                       | N                       | 9,682                | 8,375              | 13,855               | 11,028              |
|                                                | Product market fluidity | 0.001<br>(1.230)     | −0.000<br>(−0.343) | 0.007***<br>(7.512)  | 0.011***<br>(9.068) |
| WW-index                                       | N                       | 11,100               | 6,957              | 12,814               | 12,069              |
|                                                | Product market fluidity | 0.001*<br>(1.781)    | −0.000<br>(−0.031) | 0.007***<br>(8.015)  | 0.010***<br>(9.072) |
|                                                | N                       | 10,676               | 7,381              | 10,839               | 14,044              |

non-missing stock price in Compustat. In calculating this index, Size is winsorized at (the log of) \$4.5 billion, and Age is winsorized at 37 years.

**WW index:** The index is based on [Whited and Wu \(2006\)](#) and calculated as  $\{-0.091 * \text{Cash Flow}\} - \{0.062 * \text{Dividend Payer}\} + \{0.021 * \text{Long-Term Debt}\} - \{0.044 * \text{Size}\} + \{0.102 * \text{Industry Growth}\} - \{0.035 * \text{Sales Growth}\}$ , where Long-Term Debt is long term debt over assets, Industry Growth is the four-digit SIC industry sales growth rate and Sales Growth is the firm sales growth rate.

**Debt Issue:** The ratio of the change in current and long-term debt ( $\text{DLC}_t + \text{DLTT}_t - \text{DLC}_{t-1} - \text{DLTT}_{t-1}$ ) over total assets (AT).

**IPO1 through IPO5:** Indicator variable equals one if the firm went public 1, 2, 3, 4, 5 year(s) ago.

**Industry Cash Flow Volatility:** The mean standard deviation of cash flow over assets for 10 years rolling window if there are at least 3 observations at the 2-digit SIC code level.

**Industry Equity Issuance Volatility:** The standard deviation of equity issues over total assets using 5 years rolling window averaged at the two-digit SIC code level. A minimum of 3 observations are required to calculate the standard deviation.

**Compustat HHI:** Herfindahl-Hirschman Index is computed as the sum of squared market shares based on firms' sales in Compustat at the three-digit SIC code level (See [Giroud and Mueller, 2010](#)).

**Fitted HHI:** Industry concentration ratio at the three-digit SIC code level suggested by [Hoberg and Phillips \(2010a\)](#). This HHI combines Compustat data with Herfindahl data from the U.S. Commerce Department and employee data from the Bureau of Labor Statistics. This HHI covers private and public firms and is not restricted to manufacturing firms.

**Text-based (TNIC) HHI:** Industry concentration ratio at the three-digit SIC code level based on the text-based network industry classification obtained from Hoberg and Phillips data library. This HHI is based on product descriptions from annual firm 10-K filings with the SEC from 1996 to 2007 (See [Hoberg and Phillips, 2010a, b](#)).

**Product Market Fluidity:** Developed by [Hoberg et al. \(2014\)](#), fluidity captures the competitive threats faced by a firm and the changes in rival firms' products surrounding the firm. Higher fluidity indicates greater competition. The data is available from 1997 to 2007.

**Import Penetration:** Total value of imports divided by domestic absorption, defined as value of shipments minus value of exports plus value of imports. This variable is calculated at the four-digit SIC code level for manufacturing industries over 1980–2005. The data are gathered from Peter Schott's website ([Schott, 2008](#)).

**Post\_cut:** A dummy variable equals one if the tariff rate reduction has occurred in a manufacturing industry. The data are obtained from [Frésard and Valta \(2014\)](#).

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